

Stacked Ensemble Methodology: Identifying Gentrification Hotspots

Package and Data Imports + Data Cleaning

```
In [ ]: ##### IMPORT ALL NECESSARY PKGS #####
import pandas as pd
import numpy as np
import string
from fuzzywuzzy import fuzz
from sklearn.feature_extraction.text import CountVectorizer
import pyLDAvis
import pyLDAvis.lda_model

from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.tree import DecisionTreeRegressor, DecisionTreeClassifier
from xgboost import XGBRegressor, XGBClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import Lasso
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score, f1_score
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import LatentDirichletAllocation

import matplotlib.pyplot as plt

import geopandas as gpd
from shapely.wkt import loads
import plotly.express as px
import warnings
warnings.filterwarnings('ignore')

#retrieve data mined from the Google Places API (nyc_businesses1.csv) and the restaurant insp. dataset
res_path = 'datasets/outputs/nyc_businesses1.csv'
insp_path = 'datasets/nyc_open_datasets/Restaurant_Grades_20250416.csv'

business_categories = pd.read_csv(res_path)

res_insp = pd.read_csv(insp_path)

res_insp['STREET'] = res_insp['STREET'].str.lower()
res_insp['DBA'] = res_insp['DBA'].str.lower().apply(lambda s: s.translate(str.maketrans(' ', '', string.punctuation)))
res_insp['yr'] = pd.to_datetime(res_insp['GRADE DATE']).dt.year

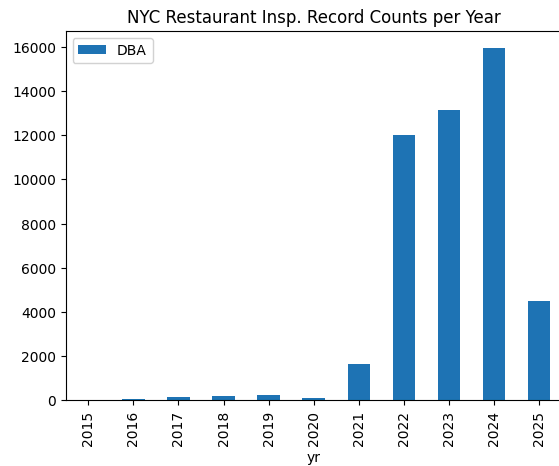
res_insp = res_insp.dropna()

res_insp['zip'] = res_insp['ZIPCODE'].astype(np.int32)
res_by_yr = res_insp[['DBA', 'zip', 'BUILDING', 'STREET', 'yr']].drop_duplicates()
res_by_yr.head(10)
```

```
Out [ ]:
```

	DBA	zip	BUILDING	STREET	yr
0	morris park bake shop	10462	1007	morris park avenue	2024
3	morris park bake shop	10462	1007	morris park avenue	2023
8	dj reynolds	10019	351	west 57 street	2025
10	dj reynolds	10019	351	west 57 street	2023
13	riviera caterers	11224	2780	stillwell avenue	2022
14	wilkens fine food	11234	7114	avenue u	2025
16	wilkens fine food	11234	7114	avenue u	2024
21	wilkens fine food	11234	7114	avenue u	2022
24	taste the tropics ice cream	11226	1839	nostrand avenue	2024
27	taste the tropics ice cream	11226	1839	nostrand avenue	2023

```
In [ ]: #note that each record represents a single code violation, meaning a restaurant can have multiple data points...
res_by_yr.groupby(['yr']).count()[['DBA']].plot(kind='bar', title='NYC Restaurant Insp. Record Counts per Year')
plt.show()
```

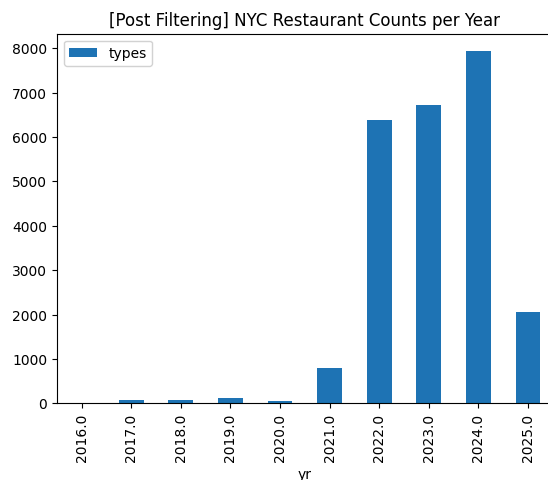


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In [ ]: #do a direct join on the zip-code and building numbers
business_categories['BUILDING'] = business_categories['formattedAddress'].str.split(' ').apply(lambda x: x[0])
business_categories['displayName'] = business_categories['displayName'].str.lower().apply(lambda s: s.translate(str.maketrans('', '', string.punctuation)))
ttl = business_categories.merge(
    res_by_yr,
    how='left',
    on=['zip', 'BUILDING']
)

ttl = ttl.dropna()

#perform a fuzzy match/join on the restaurant names...
ttl['fuzz_ratio'] = ttl.apply(lambda r: fuzz.ratio(r['displayName'], r['DBA']), axis=1)
ttl_name_parse = ttl.loc[ttl['fuzz_ratio'] > 25].sort_values(by=['fuzz_ratio'])
ttl_name_parse.head(50)

#data seems sparse prior to 2022...only consider 2022-2024
ttl_name_parse.groupby(['yr']).count()[['types']].plot(kind='bar', title='[Post Filtering] NYC Restaurant Counts per Year')
plt.show()
```



```
In [ ]: ##### DEPRECATED: Using zillow data to get median rent estimates per zip-code/year #####

zillow = pd.read_csv('datasets/zillow/Zip_zori_uc_sfrcondomfr_sm_month.csv')
zillow_nyc = zillow.loc[zillow['City']=='New York']
rent_monthly_p = zillow_nyc[['RegionName']] + [
    i for i in zillow_nyc.columns
```

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        if len(i.split('-')) == 3
            and int(i.split('-')[0]) >= 2014
            and int(i.split('-')[0]) < 2025
    ]
]

rent_monthly_tidy = pd.melt(rent_monthly_p, id_vars='RegionName')
rent_monthly_tidy['year'] = rent_monthly_tidy['variable'].str.split('-').apply(lambda x:x[0])
rent_monthly_tidy = rent_monthly_tidy\
    .ffill()\
    .drop(columns=['variable'])\
    .rename(columns={
        'RegionName':'zip_code',
        'value':'rent'
    })\
    .groupby(['zip_code','year']).mean()\
    .reset_index()

rent_monthly_tidy['zip_code'] = rent_monthly_tidy['zip_code'].astype(str)
rent_nyc_pivot = rent_monthly_tidy.pivot(
    index='zip_code', columns='year', values='rent'
)

rent_change_pivot_zillow = rent_nyc_pivot.div(rent_nyc_pivot['2015'],axis=0)-1
rent_change_pivot_zillow.head(10)

```

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Out[ ]:
   year  2015   2016   2017   2018   2019   2020   2021   2022   2023   2024
zip_code
10001  0.0  0.006006 -0.013769 -0.000709  0.036649 -0.042370  0.010997  0.205286  0.231849  0.268840
10002  0.0  0.017578  0.031882  0.053497  0.081383  0.016909  0.028776  0.244991  0.264129  0.286204
10003  0.0  0.018181  0.028188  0.044347  0.072727  0.007582  0.024290  0.256503  0.299623  0.340851
10004  0.0 -0.121405 -0.148968 -0.141138 -0.109580 -0.183109 -0.154772  0.044439  0.064973  0.123936
10005  0.0  0.002534 -0.005651  0.025975  0.056846  0.007947  0.019859  0.217989  0.253319  0.291088
10006  0.0 -0.121405 -0.147391 -0.154905 -0.094440 -0.138905 -0.135635  0.058432  0.089164  0.129415
10007  0.0  0.002534  0.468077  0.486518  0.552854  0.485547  0.546448  0.784083  0.869954  0.953419
10009  0.0  0.021085  0.033979  0.049052  0.078888  0.003042  0.024139  0.257914  0.300074  0.326688
10010  0.0  0.014159  0.016996  0.025274  0.051566 -0.022283 -0.006308  0.199299  0.245633  0.285011
10011  0.0  0.014771  0.015642  0.032588  0.056759 -0.085621  0.019546  0.224453  0.266396  0.318582

```

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In [ ]: ##### Clean the ACS Dataset to obtain rent data #####

acs = pd.read_csv('datasets/ACS_update.csv')
acs = acs.replace(-666666666,np.nan).dropna()
acs_2015 = acs.loc[acs['year']>=2015]

#find a subset of features to train model on...
feats = [
    'less_than_9th_grade',
    '9th_to_12th_grade',
    'high_school_graduate',
    'some_college_no_degree',
    'associates_degree',
    'bachelors_degree',
    'graduate_or_professional_degree',
    'pct_children_teens_5-19',
    'pct_young_adults_20-34',
    'pct_middle_aged_35-59',
    'pct_older_adults_60+',
    'pct_1_person',
    'pct_2_person',
    'pct_3_person',
    'pct_4_or_more_person',
    'household_median_income_1yr',
    'median_rent'
]

#calculate growth indices for each variable...
feats_update = feats.copy()
for f in feats:
    f_name_new = f+'_change'
    acs_2015[f_name_new] = (acs_2015[f]/acs_2015.groupby(['zcta'])[f].transform('last'))-1
    feats_update.append(f_name_new)

acs_2015_tr = acs_2015[['zcta','year']+feats_update].rename(columns={'zcta':'zip_code'})
acs_2015_tr['zip_code'] = acs_2015_tr['zip_code'].astype(str)

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acs_2015_tr['year'] = acs_2015_tr['year'].astype(str)

acs_2015_tr.head(10)

#build a pivot table of rents akin to the zillow data above...
rent_change_pivot = acs_2015_tr.pivot(
    index='zip_code', columns='year', values='median_rent_change'
)

rent_change_pivot

```

```

Out[ ]:
   year 2015  2016  2017  2018  2019  2020  2021  2022  2023
zip_code
10001  0.0  0.062846 -0.038713  0.038210  0.069884  0.090498  0.140774  0.319759  0.520362
10002  0.0  0.041950  0.061224  0.091837  0.111111  0.109977  0.187075  0.287982  0.368481
10003  0.0  0.042857  0.084236  0.132020  0.148276  0.200493  0.220690  0.345320  0.365517
10004  0.0  0.021045  0.069449  0.134690  0.165907  0.190109  0.227990  0.227990  0.227990
10005  0.0  0.029815  0.076422  0.105552  0.150103  0.199794  0.199794  0.199794  0.199794
...     ...     ...     ...     ...     ...     ...     ...     ...
11691  0.0  0.013966  0.040968  0.117318  0.149907  0.189944  0.255121  0.325885  0.344507
11692  0.0  0.023613  0.122786  0.192444  0.305785  0.381346  0.495868  0.540732  0.504132
11693  0.0 -0.011765  0.081283  0.109091  0.132620  0.031016  0.112299  0.216043  0.281283
11694  0.0  0.031353  0.070132  0.089934  0.136139  0.183993  0.284653  0.392739  0.474422
11697  0.0      NaN      NaN      NaN -1.000001 -1.000001 -1.000001      NaN      NaN

```

181 rows × 9 columns

Feature Mining: Restaurant/Business Data

```

In [ ]: #identify a restaurant's active status based on the inspection date
ttl_name_parse['first_appearance'] = ttl_name_parse.groupby(['displayName'])['yr'].transform(np.min)
ttl_name_parse['latest_appearance'] = ttl_name_parse.groupby(['displayName'])['yr'].transform(lambda x: min(2024,np.max(x)))

TOTAL = []

#only look at 2022-2023 to align with the other datasets for the ensemble...
yrs = [2022,2023]

#apply a countvectorizer object on each restaurant dataset per year (spec. the restaurant type flags), combined w rent info from ACS
for yr in yrs:
    sample = \
        ttl_name_parse\
            .loc[
                (ttl_name_parse['first_appearance']<=yr)
                &(ttl_name_parse['latest_appearance']>=yr)
            ]\
            [['types',
              'zip',
              'priceLevel',
              'displayName',
              ]].drop_duplicates()
    sample['zip'] = sample['zip'].astype(str)

    cv = CountVectorizer()

    #fit the preprocessor on the 'types' column (comma sep. strings of type descriptor flags)
    vectorized = cv.fit_transform(sample['types']).toarray()
    features = cv.get_feature_names_out()
    cat_analysis_r = \
        pd.DataFrame(
            vectorized,
            columns=features,
        )
    cat_analysis_r['zip'] = sample.reset_index()['zip']
    cat_analysis = cat_analysis_r.groupby(['zip']).sum()

    cat_analysis.columns = [f'ttl_{i}' for i in cat_analysis.columns]

#combined w rent data
pricing = sample\
    .groupby(['priceLevel','zip'])['displayName']\
    .count()\
    .reset_index()\
    .pivot(index='zip',columns='priceLevel',values='displayName')\

```

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    .drop(columns='FREE')\
    .fillna(0)

rent = rent_nyc_pivot[[str(yr)]].rename(columns={str(yr): 'rent'})
rent_change = rent_change_pivot[[str(yr)]].rename(columns={str(yr): 'rent_change'})
by_yr = pd.concat((cat_analysis, pricing, rent, rent_change), axis=1)
by_yr['yr'] = yr
TOTAL.append(by_yr.dropna())

TOTAL = pd.concat(TOTAL).dropna(axis=1)

#establish our prediction variable based on the rent_change target
TOTAL['is_gentrified'] = TOTAL['rent_change'] > 0.343 #see literature
TOTAL.head(25)

```

```

Out [ ]:
    ttl_acai_shop  ttl_afghani_restaurant  ttl_african_restaurant  ttl_american_restaurant  ttl_amusement_center  ttl_art_gallery  ttl_asian_grocery_store  ttl_asian_restaurant  ttl_athletic_field  ttl_auditorium  ...  ttl_wholesaler  ttl_wine_bar  ttl_yoga_studio
10001           0.0                0.0                0.0                20.0                0.0                0.0                0.0                10.0                0.0                0.0  ...                0.0                8.0                0.0
10002           0.0                0.0                0.0                12.0                0.0                0.0                0.0                10.0                0.0                0.0  ...                2.0                10.0                0.0
10003           1.0                0.0                0.0                15.0                0.0                0.0                1.0                12.0                0.0                2.0  ...                1.0                7.0                1.0
10004           0.0                0.0                0.0                3.0                0.0                0.0                0.0                0.0                0.0                0.0  ...                0.0                1.0                0.0
10005           0.0                0.0                0.0                4.0                0.0                0.0                0.0                0.0                0.0                0.0  ...                0.0                1.0                0.0
10006           0.0                0.0                0.0                3.0                0.0                0.0                0.0                0.0                0.0                0.0  ...                0.0                0.0                0.0
10007           0.0                0.0                0.0                4.0                0.0                0.0                0.0                1.0                0.0                0.0  ...                0.0                0.0                0.0
10009           0.0                0.0                0.0                11.0                0.0                1.0                0.0                7.0                0.0                0.0  ...                0.0                13.0                0.0
10010           0.0                0.0                0.0                13.0                0.0                0.0                0.0                2.0                0.0                1.0  ...                0.0                2.0                0.0
10011           0.0                0.0                0.0                16.0                0.0                1.0                0.0                7.0                0.0                0.0  ...                4.0                10.0                0.0
10012           0.0                0.0                0.0                11.0                0.0                0.0                0.0                3.0                0.0                0.0  ...                2.0                6.0                0.0
10013           0.0                0.0                0.0                18.0                0.0                0.0                0.0                17.0                0.0                0.0  ...                1.0                13.0                0.0
10014           0.0                0.0                0.0                24.0                0.0                0.0                0.0                5.0                0.0                0.0  ...                2.0                10.0                0.0
10016           0.0                0.0                0.0                14.0                0.0                0.0                0.0                8.0                0.0                0.0  ...                2.0                6.0                0.0
10017           0.0                0.0                0.0                18.0                0.0                0.0                1.0                6.0                0.0                0.0  ...                0.0                2.0                0.0
10018           0.0                0.0                0.0                19.0                0.0                0.0                0.0                6.0                0.0                0.0  ...                2.0                3.0                0.0
10019           0.0                1.0                0.0                35.0                0.0                0.0                0.0                4.0                0.0                0.0  ...                2.0                11.0                0.0
10021           0.0                0.0                0.0                8.0                0.0                0.0                0.0                3.0                0.0                0.0  ...                0.0                2.0                0.0
10022           1.0                0.0                0.0                31.0                0.0                0.0                0.0                10.0                0.0                0.0  ...                0.0                5.0                0.0
10023           0.0                0.0                0.0                2.0                0.0                0.0                0.0                2.0                0.0                0.0  ...                0.0                2.0                0.0
10024           0.0                0.0                0.0                12.0                0.0                0.0                0.0                2.0                0.0                0.0  ...                1.0                3.0                0.0
10025           0.0                0.0                1.0                6.0                0.0                0.0                0.0                3.0                0.0                0.0  ...                0.0                2.0                0.0
10026           0.0                0.0                1.0                2.0                0.0                0.0                0.0                1.0                0.0                0.0  ...                0.0                1.0                0.0
10027           0.0                0.0                0.0                6.0                0.0                0.0                0.0                0.0                0.0                0.0  ...                0.0                0.0                0.0
10028           0.0                0.0                0.0                9.0                0.0                0.0                0.0                2.0                0.0                0.0  ...                0.0                4.0                0.0

```

25 rows x 135 columns

Dimension Reduction + Topic Modeling

```

In [ ]: #remove generic flags
cols = [i for i in TOTAL.columns if 'ttl_' in i]
cols.remove('ttl_food_store')
cols.remove('ttl_store')
cols.remove('ttl_food_delivery')
cols.remove('ttl_meal_delivery')

#for the ldaviz to work, need to reconstruct the document matrix with a singularly defined countvectorizer object
v=np.ones((TOTAL.shape[0], len(cols)),dtype=object)
for idx,f in enumerate(cols):
    v[:,idx] = TOTAL[f].apply(lambda x: int(x)*f'{f} ').to_numpy()

v=[' '.join(i) for i in v]

#apply LDA...try 6 topics
cv_analysis = CountVectorizer()
lda = LatentDirichletAllocation(n_components=6, )

doc_matrix = cv_analysis.fit_transform(v)

```

```
lda.fit(doc_matrix)

#print most relevant type flags per topic...
topic_components = lda.components_
topics = dict()
for n,t in enumerate(topic_components):
    idx = t.argsort()[-5:]
    items = [cols[i] for i in idx]
    topics[f'TOPIC {n}'] = items

topics
```

```
Out [ ]: {'TOPIC 0': ['ttl_meal_takeaway',
                    'ttl_cafe',
                    'ttl_italian_restaurant',
                    'ttl_american_restaurant',
                    'ttl_bar'],
         'TOPIC 1': ['ttl_cafe',
                    'ttl_event_venue',
                    'ttl_night_club',
                    'ttl_american_restaurant',
                    'ttl_bar'],
         'TOPIC 2': ['ttl_japanese_restaurant',
                    'ttl_cafe',
                    'ttl_asian_restaurant',
                    'ttl_bakery',
                    'ttl_chinese_restaurant'],
         'TOPIC 3': ['ttl_bar',
                    'ttl_bakery',
                    'ttl_chinese_restaurant',
                    'ttl_mexican_restaurant',
                    'ttl_pizza_restaurant'],
         'TOPIC 4': ['ttl_market',
                    'ttl_convenience_store',
                    'ttl_grocery_store',
                    'ttl_pizza_restaurant',
                    'ttl_italian_restaurant'],
         'TOPIC 5': ['ttl_dessert_shop',
                    'ttl_coffee_shop',
                    'ttl_confectionery',
                    'ttl_cafe',
                    'ttl_bar']}
```

```
In [ ]: #create LDAviz
vis_data = pyLDAvis.lda_model.prepare(lda, doc_matrix, vectorizer = cv_analisis, sort_topics=False)

# Visualize the LDA model using pyLDAvis in a Jupyter Notebook
pyLDAvis.enable_notebook()
pyLDAvis.display(vis_data)
pyLDAvis.save_html(vis_data, 'datasets/outputs/lda.html')
```

```
In [ ]: #transform the topic/word-level data into something useful geo-spatially....
#assess each topic's relevance per zip-code in NYC

TOTAL_w_LDA = TOTAL.copy()
tps = range(1,7)
for tp in tps:
    sv = vis_data.sorted_terms(topic=tp, _lambda=0.30).set_index('Term')['relevance']
    items = sv.index
    top_10_relevant = ','.join(list(items[:10]))

    print(f'**** TOPIC {tp}: MOST RELEVANT TYPES: {top_10_relevant} ****')

    TOTAL_w_LDA[f'LDArelevance{tp}'] = TOTAL_w_LDA[items].values @ sv['relevance'].to_numpy().reshape(-1,1)

ldaonly = TOTAL_w_LDA[[f'LDArelevance{tp}' for tp in tps]]
ldaonly.index.name = 'zip'
ldaonly = ldaonly.reset_index().groupby(['zip']).sum()
lda_temp = ldaonly.copy()

#calculate a most/least likely feature...
ldaonly['most_likely'] = lda_temp.idxmax(axis=1).str[-1]
ldaonly['least_likely'] = lda_temp.idxmin(axis=1).str[-1]

warnings.filterwarnings('ignore')

#get shapefile for nyc zctas
geo_zip_nyc = gpd.GeoDataFrame(pd.read_csv('datasets/nyc_open_datasets/Modified_Zip_Code_Tabulation_Areas_MODZCTA_20250421.csv'))
geo_zip_nyc['zip'] = geo_zip_nyc['MODZCTA'].astype(str)

geo_zip_nyc = geo_zip_nyc\
    .rename(columns={
```

```

        'the_geom': 'geometry'
    })\
    .set_index('zip')[['geometry']]

geo_zip_nyc['geometry'] = geo_zip_nyc['geometry'].apply(loads)

w_geo = gpd.GeoDataFrame(pd.concat((geo_zip_nyc, ldaonly), axis=1).dropna(), crs="EPSG:4326")
fig = px.choropleth_mapbox(
    w_geo,
    geojson=w_geo['geometry'],
    locations=w_geo.index,
    color="most_likely",
    featureidkey="id",
    center={"lat": w_geo.geometry.centroid.y.mean(), "lon": w_geo.geometry.centroid.x.mean()},
    zoom=10,
    mapbox_style="carto-positron",
)
fig.update_layout(margin={"r": 0, "t": 0, "l": 0, "b": 0})
fig.show()

fig = px.choropleth_mapbox(
    w_geo,
    geojson=w_geo['geometry'],
    locations=w_geo.index,
    color="least_likely",
    featureidkey="id",
    center={"lat": w_geo.geometry.centroid.y.mean(), "lon": w_geo.geometry.centroid.x.mean()},
    zoom=10,
    mapbox_style="carto-positron",
)
fig.update_layout(margin={"r": 0, "t": 0, "l": 0, "b": 0})
#fig.show()

**** TOPIC 1 MOST RELEVANT TYPES: ttl_italian_restaurant,ttl_fine_dining_restaurant,ttl_brunch_restaurant,ttl_greek_restaurant,ttl_american_restaurant,ttl_bar,ttl_french_restaurant,ttl_japanese_restaurant,ttl_sushi_restaurant,ttl_meal_takeaway ****
**** TOPIC 2 MOST RELEVANT TYPES: ttl_bar,ttl_korean_restaurant,ttl_night_club,ttl_event_venue,ttl_karaoke,ttl_sports_complex,ttl_sports_activity_location,ttl_bar_and_grill,ttl_sports_club,ttl_wine_bar ****
**** TOPIC 3 MOST RELEVANT TYPES: ttl_chinese_restaurant,ttl_asian_restaurant,ttl_bakery,ttl_tea_house,ttl_thai_restaurant,ttl_barbecue_restaurant,ttl_barbecue_area,ttl_vietnamese_restaurant,ttl_japanese_restaurant,ttl_dessert_restaurant ****
**** TOPIC 4 MOST RELEVANT TYPES: ttl_mexican_restaurant,ttl_pizza_restaurant,ttl_african_restaurant,ttl_chinese_restaurant,ttl_bakery,ttl_breakfast_restaurant,ttl_fast_food_restaurant,ttl_juice_shop,ttl_deli,ttl_meal_takeaway ***
*
**** TOPIC 5 MOST RELEVANT TYPES: ttl_convenience_store,ttl_market,ttl_grocery_store,ttl_insurance_agency,ttl_extended_stay_hotel,ttl_resort_hotel,ttl_place_of_worship,ttl_yoga_studio,ttl_consultant,ttl_lawyer ****
**** TOPIC 6 MOST RELEVANT TYPES: ttl_dessert_shop,ttl_confectionery,ttl_ice_cream_shop,ttl_cafe,ttl_coffee_shop,ttl_park,ttl_garden,ttl_wholesaler,ttl_bar,ttl_chocolate_shop ****

```

In []: ldaonly.reset_index()

Out []:

	zip	LDArelevance1	LDArelevance2	LDArelevance3	LDArelevance4	LDArelevance5	LDArelevance6	most_likely	least_likely
0	10001	-480.41209	-526.14753	-588.22565	-721.44652	-1174.59185	-591.49913	1	5
1	10002	-498.89410	-623.37413	-594.09106	-703.28944	-1175.61132	-604.30038	1	5
2	10003	-728.19423	-896.94534	-870.05936	-966.88446	-1631.56341	-847.22019	1	5
3	10004	-104.81832	-125.57739	-120.83138	-103.84597	-212.59478	-123.29044	4	5
4	10005	-42.20754	-28.29182	-48.82858	-35.74420	-57.75048	-46.13184	2	5
...	...	...	...	...	...	...	...	...	...
134	11432	-24.74311	-36.61472	-24.41635	-29.07830	-39.53289	-35.62413	3	5
135	11434	-14.94732	-24.16593	-8.11123	-15.14979	-13.93657	-25.54256	3	6
136	11435	-24.61804	-44.83961	-27.23232	-23.99234	-52.23770	-37.93268	4	5
137	11691	-6.94824	-12.18899	-7.02465	-9.76012	0.00000	-14.75670	5	6
138	11694	-21.46455	-24.45647	-28.66249	-25.21619	-35.77841	-25.21173	1	5

139 rows x 9 columns

Modeling: Business Dataset

```

In [ ]: #TOTAL = TOTAL.reset_index(names='zip_code')
train,test = TOTAL.loc[TOTAL['yr'].isin([2022])],TOTAL.loc[TOTAL['yr']==2023]

training_feats = [
    'ttl_night_club',
    'ttl_greek_restaurant',
    'ttl_mediterranean_restaurant',
    'ttl_seafood_restaurant',
    'ttl_bar',
    'ttl_pizza_restaurant',
    'ttl_chinese_restaurant',
    'ttl_meal_delivery',
    'ttl_bakery',

```

```

'ttl_american_restaurant',
'ttl_coffee_shop',
'ttl_dessert_shop',
'EXPENSIVE',
'MODERATE',
'INEXPENSIVE'
]

prediction_var = 'is_gentrified'

```

```
In [ ]: train['is_gentrified'].value_counts()
```

```
Out[ ]: is_gentrified
True    71
False   67
Name: count, dtype: int64
```

```
In [ ]: ##### Build helper functions for training/cross-validation #####
#use this code as a workflow template for the project group
```

```

def run_prediction(
    model,
    train,
    test,
    training_feats,
    prediction_var,
    params_config,
    nm,
    plot=False
):
    """
    Performs a Grid Search on a set of parameters, finds an optimal model, and makes probabilistic predictions.
    Assumes a binary classifier.

    Inputs:
    **model: a sklearn or XGBoost model object
    **train: the training dataset
    **test: the test dataset to make predictions on
    **training_feats: the features you want to train the classifier on
    **prediction_var: the variable we want to predict. should be boolean in nature
    **params_config: a dictionary of hyperparameters to test the model
    **nm: the name of the model

    Outputs: A tuple consisting of the following:
    ** the probabilities of class assignment (p(Y=0) and p(Y=1)) per item in the test set
    ** the feature importance scores associated with each trained model
    ** the best model after the gridsearch

    """

    #build the training/prediction data
    Xtrain,ytrain = train[training_feats].values, train[prediction_var]
    Xtest,ytest = test[training_feats].values, test[prediction_var]

    #grid search CV
    model_cv = GridSearchCV(estimator=model, param_grid=params_config, cv=3)

    #fits/predicts the optimized model
    model_cv.fit(Xtrain, ytrain)

    predictions = model_cv.predict(Xtest)
    train['predictions'] = model_cv.predict_proba(Xtrain)[:,:1]
    test['predictions'] = model_cv.predict_proba(Xtest)[:,:1]

    corr = np.corrcoef(test['rent_change'], test['predictions'])[0][1]
    if plot:
        _, ax = plt.subplots()
        ax.scatter(
            test.loc[test[prediction_var]]['predictions'],
            test.loc[test[prediction_var]]['rent_change'],
            label='Gentrified'
        )
        ax.scatter(
            test.loc[~test[prediction_var]]['predictions'],
            test.loc[~test[prediction_var]]['rent_change'],
            label='Not Gentrified'
        )
        ax.axvline(0.5, label='Classifier Decision Boundary', color='red')
        ax.set_title(f'{nm}: Corr[Gentrification Score, Rent Change]={corr}')
        ax.set_xlabel('Probability (Gentrification Hotspot Score)')
        ax.set_ylabel('Rent Pct Change')
        ax.legend()
        plt.show()

```

```

# print out various metrics (accuracy + F1 -- weighted prec. and recall)
print(f'*** {nm} Accuracy: {accuracy_score(ytest, predictions)} ***')
print(f'*** {nm} F1: {f1_score(ytest, predictions)} ***')

if nm == '[BASELINE] NaiveBayes':
    return train, test, {}, model_cv.best_estimator_, corr

# gather info on the importance of each feature (20 top feats)...
feature_importances = model_cv.best_estimator_.feature_importances_
idx_sort_topfeats = np.argsort(feature_importances)[::-1][:20]
feat_importance_scores = {training_feats[i]: feature_importances[i] for i in idx_sort_topfeats}

return train, test, feat_importance_scores, model_cv.best_estimator_, corr

# build gridsearch configs
def run_all_models(train, test, training_feats, plot=True, target='is_gentrified'):
    models = [
        {
            'model': GaussianNB(),
            'params_config': {},
            'nm': '[BASELINE] NaiveBayes'
        },
        {
            'model': DecisionTreeClassifier(),
            'params_config': {
                'max_depth': [3, 5, 7],
                'min_samples_split': [2, 5, 10],
                'min_samples_leaf': [1, 2, 5]
            },
            'nm': 'Single Decision Tree'
        },
        {
            'model': RandomForestClassifier(),
            'params_config': {
                'n_estimators': [50, 100, 200, 300],
                'max_depth': [3, 5, 7],
                'min_samples_split': [2, 5, 10],
                'min_samples_leaf': [1, 2, 5]
            },
            'nm': 'Random Forests'
        },
        {
            'model': XGBClassifier(),
            'params_config': {
                'n_estimators': [50, 100, 200, 300],
                'max_depth': [3, 5, 7, 9],
                'learning_rate': [0.01, 0.05, 0.1, 0.2]
            },
            'nm': 'XGBoost'
        }
    ]

    # run the predictions, and record in the above list ('models')
    for m_config in models:
        m_config['training_feats'] = training_feats
        m_config['prediction_var'] = target
        m_config['plot'] = plot

        # run model, unpack deliverables
        in_sample, out_sample, fi, m, perf = run_prediction(train=train.copy(), test=test.copy(), **m_config)

        # print feature importance scores
        print(f'*** Feature Importance Scores *** \n {fi}')

        # record the results per dict in the list
        m_config['feature_importances'] = fi
        m_config['train'] = in_sample
        m_config['test'] = out_sample
        m_config['model'] = m
        m_config['perf'] = perf

    return models

# TOTAL = TOTAL.reset_index(names='zip_code')
train, test = TOTAL.loc[TOTAL['yr'].isin([2022])], TOTAL.loc[TOTAL['yr']==2023]

training_feats = [
    'ttl_night_club',
    'ttl_greek_restaurant',
    'ttl_mediterranean_restaurant',
    'ttl_seafood_restaurant',

```

```

'ttl_bar',
'ttl_pizza_restaurant',
'ttl_chinese_restaurant',
'ttl_meal_delivery',
'ttl_bakery',
'ttl_american_restaurant',
'ttl_coffee_shop',
'ttl_dessert_shop',
'EXPENSIVE',
'MODERATE',
'INEXPENSIVE'
]

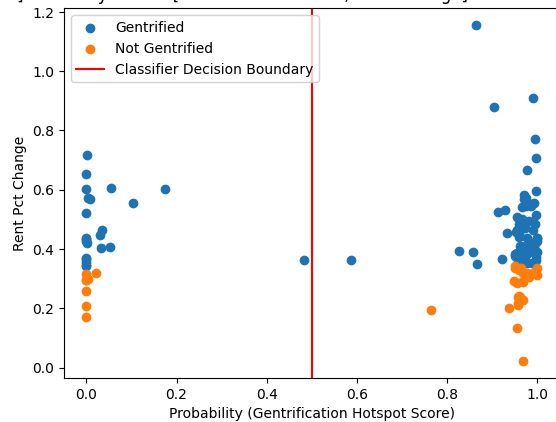
prediction_var = 'is_gentrified'

business_models = run_all_models(
    train, test, training_feats, plot=True
)

business_models

```

[BASELINE] NaiveBayes: Corr[Gentrification Score, Rent Change]=-0.019744396343331215

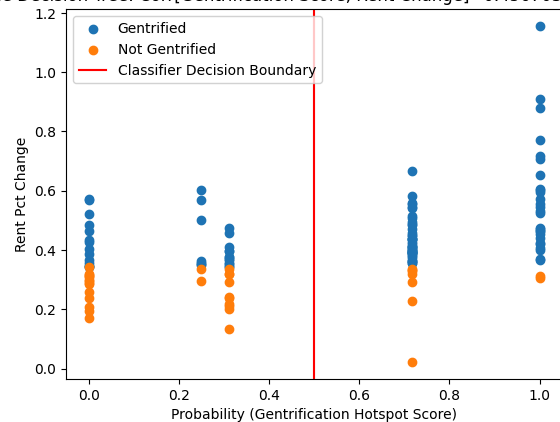


```

*** [BASELINE] NaiveBayes Accuracy: 0.6402877697841727 ***
*** [BASELINE] NaiveBayes F1: 0.7641509433962265 ***
*** Feature Importance Scores ***
{}

```

Single Decision Tree: Corr[Gentrification Score, Rent Change]=0.4567037859017066

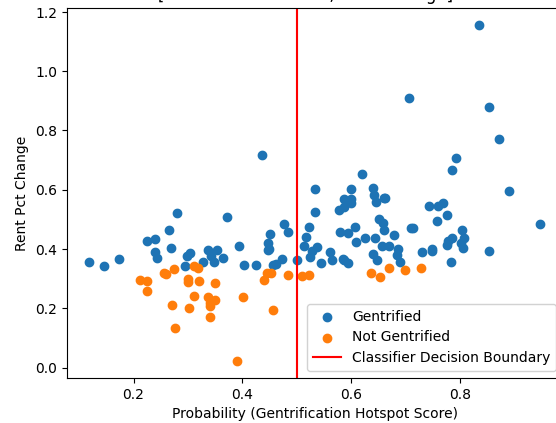


```

*** Single Decision Tree Accuracy: 0.6906474820143885 ***
*** Single Decision Tree F1: 0.7650273224043715 ***
*** Feature Importance Scores ***
{'ttl_coffee_shop': 0.3084095798814296, 'ttl_bakery': 0.208503017914835, 'EXPENSIVE': 0.171499524783321, 'INEXPENSIVE': 0.10931077336270391, 'ttl_american_restaurant': 0.08480767401110546, 'ttl_meal_delivery': 0.0618847367746120, 'ttl_dessert_shop': 0.055584693271993064, 'MODERATE': 0.0, 'ttl_chinese_restaurant': 0.0, 'ttl_pizza_restaurant': 0.0, 'ttl_bar': 0.0, 'ttl_seafood_restaurant': 0.0, 'ttl_mediterranean_restaurant': 0.0, 'ttl_greek_restaurant': 0.0, 'ttl_night_club': 0.0}

```

Random Forests: Corr[Gentrification Score, Rent Change]=0.5307895273396422



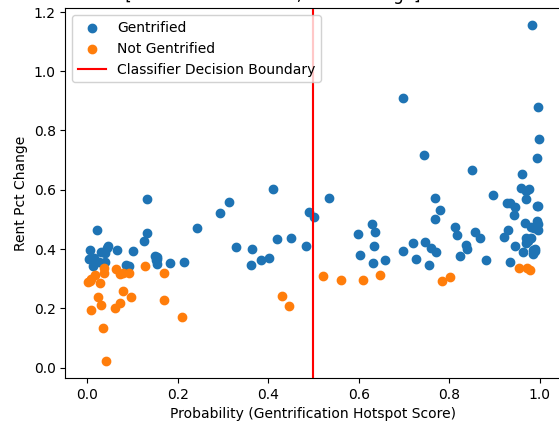
*** Random Forests Accuracy: 0.7122302158273381 ***

*** Random Forests F1: 0.7802197802197802 ***

*** Feature Importance Scores ***

{'INEXPENSIVE': 0.11843125037276991, 'ttl_bakery': 0.0986381679265946, 'MODERATE': 0.09358974317639111, 'ttl_coffee_shop': 0.07885647052184007, 'ttl_meal_delivery': 0.07282037509745128, 'ttl_bar': 0.07267888080624399, 'EXPENSIVE': 0.0679522810404613, 'ttl_american_restaurant': 0.06181490392631924, 'ttl_mediterranean_restaurant': 0.06136420146167838, 'ttl_chinese_restaurant': 0.0567049804113039, 'ttl_seafood_restaurant': 0.050512951145514996, 'ttl_pizza_restaurant': 0.04616568671477388, 'ttl_dessert_shop': 0.04506534586209119, 'ttl_night_club': 0.039567918616381405, 'ttl_greek_restaurant': 0.035836842920184506}

XGBoost: Corr[Gentrification Score, Rent Change]=0.5190943638301379



*** XGBoost Accuracy: 0.6546762589928058 ***

*** XGBoost F1: 0.7303370786516854 ***

*** Feature Importance Scores ***

{'EXPENSIVE': 0.14362577, 'ttl_greek_restaurant': 0.10716407, 'ttl_coffee_shop': 0.10611808, 'ttl_bakery': 0.09289351, 'ttl_dessert_shop': 0.072459035, 'ttl_seafood_restaurant': 0.07061396, 'ttl_american_restaurant': 0.06258531, 'MODERATE': 0.05700505, 'ttl_night_club': 0.045059413, 'ttl_chinese_restaurant': 0.043990742, 'ttl_meal_delivery': 0.04184405, 'INEXPENSIVE': 0.041728314, 'ttl_pizza_restaurant': 0.039535638, 'ttl_bar': 0.03785461, 'ttl_mediterranean_restaurant': 0.037522454}

```

Out[ ]: [{'model': GaussianNB(),
        'params_config': {},
        'nm': '[BASELINE] NaiveBayes',
        'training_feats': ['ttl_night_club',
                           'ttl_greek_restaurant',
                           'ttl_mediterranean_restaurant',
                           'ttl_seafood_restaurant',
                           'ttl_bar',
                           'ttl_pizza_restaurant',
                           'ttl_chinese_restaurant',
                           'ttl_meal_delivery',
                           'ttl_bakery',
                           'ttl_american_restaurant',
                           'ttl_coffee_shop',
                           'ttl_dessert_shop',
                           'EXPENSIVE',
                           'MODERATE',
                           'INEXPENSIVE'],
        'prediction_var': 'is_gentrified',
        'plot': True,
        'feature_importances': {},
        'train':
            ttl_acai_shop    ttl_afghani_restaurant    ttl_african_restaurant \
10001          0.0          0.0          0.0
10002          0.0          0.0          0.0
10003          1.0          0.0          0.0
10004          0.0          0.0          0.0
10005          0.0          0.0          0.0
...
11432          0.0          0.0          0.0
11434          0.0          0.0          1.0
11435          0.0          0.0          1.0
11691          0.0          0.0          0.0
11694          0.0          0.0          0.0

            ttl_american_restaurant    ttl_amusement_center    ttl_art_gallery \
10001          20.0          0.0          0.0
10002          12.0          0.0          0.0
10003          15.0          0.0          0.0
10004          3.0          0.0          0.0
10005          4.0          0.0          0.0
...
11432          0.0          0.0          0.0
11434          0.0          0.0          0.0
11435          1.0          0.0          0.0
11691          1.0          0.0          0.0
11694          0.0          0.0          0.0

            ttl_asian_grocery_store    ttl_asian_restaurant    ttl_athletic_field \
10001          0.0          10.0          0.0
10002          0.0          10.0          0.0
10003          1.0          12.0          0.0
10004          0.0          0.0          0.0
10005          0.0          0.0          0.0
...
11432          0.0          0.0          0.0
11434          0.0          0.0          0.0
11435          0.0          0.0          0.0
11691          0.0          0.0          0.0
11694          0.0          0.0          0.0

            ttl_auditorium    ...    ttl_wine_bar    ttl_yoga_studio    EXPENSIVE \
10001          0.0    ...          8.0          0.0          17.0
10002          0.0    ...         10.0          0.0          12.0
10003          2.0    ...          7.0          1.0          25.0
10004          0.0    ...          1.0          0.0          2.0
10005          0.0    ...          1.0          0.0          4.0
...
11432          0.0    ...          0.0          0.0          0.0
11434          0.0    ...          0.0          0.0          0.0
11435          0.0    ...          0.0          0.0          0.0
11691          0.0    ...          0.0          0.0          0.0
11694          0.0    ...          0.0          0.0          0.0

            INEXPENSIVE    MODERATE    rent    rent_change    yr    is_gentrified \
10001          25.0          90.0    4870.677598    0.319759    2022    False
10002          62.0          75.0    3554.796326    0.287982    2022    False
10003          68.0          108.0    3941.807689    0.345320    2022    True
10004          13.0          11.0    3946.666521    0.227990    2022    False
10005          1.0          5.0    4275.182475    0.199794    2022    False
...
11432          7.0          2.0    2142.269355    0.342445    2022    False
11434          4.0          3.0    2980.772466    0.211454    2022    False
11435          6.0          3.0    1846.797387    0.337604    2022    False
11691          2.0          1.0    2315.852443    0.325885    2022    False

```

```

11694      2.0      4.0 3886.183980      0.392739 2022      True

      predictions
10001 6.021692e-40
10002 2.213347e-30
10003 4.292842e-15
10004 9.735259e-01
10005 9.583431e-01
...
11432 9.586453e-01
11434 9.532433e-01
11435 9.657839e-01
11691 9.584216e-01
11694 9.651621e-01

[138 rows x 136 columns],
'test':      ttl_acai_shop  ttl_afghani_restaurant  ttl_african_restaurant \
10001      0.0      0.0      0.0
10002      0.0      0.0      0.0
10003      1.0      0.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...
11432      0.0      0.0      0.0
11434      0.0      0.0      1.0
11435      0.0      0.0      1.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

      ttl_american_restaurant  ttl_amusement_center  ttl_art_gallery \
10001      22.0      0.0      0.0
10002      15.0      0.0      0.0
10003      17.0      0.0      0.0
10004      5.0      0.0      0.0
10005      4.0      0.0      0.0
...
11432      0.0      0.0      0.0
11434      0.0      0.0      0.0
11435      1.0      0.0      0.0
11691      1.0      0.0      0.0
11694      2.0      0.0      0.0

      ttl_asian_grocery_store  ttl_asian_restaurant  ttl_athletic_field \
10001      0.0      12.0      0.0
10002      0.0      16.0      0.0
10003      1.0      22.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...
11432      0.0      1.0      0.0
11434      0.0      0.0      0.0
11435      0.0      0.0      0.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

      ttl_auditorium ...  ttl_wine_bar  ttl_yoga_studio  EXPENSIVE \
10001      0.0 ...      12.0      0.0      18.0
10002      1.0 ...      12.0      0.0      13.0
10003      2.0 ...      9.0      1.0      27.0
10004      0.0 ...      1.0      0.0      3.0
10005      0.0 ...      1.0      0.0      4.0
...
11432      0.0 ...      0.0      0.0      0.0
11434      0.0 ...      0.0      0.0      0.0
11435      0.0 ...      0.0      0.0      0.0
11691      0.0 ...      0.0      0.0      0.0
11694      0.0 ...      0.0      0.0      0.0

      INEXPENSIVE  MODERATE      rent  rent_change  yr  is_gentrified \
10001      36.0      113.0 4978.021561 0.520362 2023      True
10002      73.0      105.0 3609.439689 0.368481 2023      True
10003      80.0      156.0 4077.081605 0.365517 2023      True
10004      13.0      14.0 4024.260110 0.227990 2023      False
10005      2.0      5.0 4399.192585 0.199794 2023      False
...
11432      12.0      2.0 2309.792636 0.397874 2023      True
11434      5.0      4.0 3135.304935 0.241410 2023      False
11435      7.0      3.0 1954.208071 0.371515 2023      True
11691      3.0      1.0 2271.818549 0.344507 2023      True
11694      2.0      6.0 4072.959038 0.474422 2023      True

      predictions
10001 1.097614e-45
10002 3.829274e-42

```

```

10003 1.748331e-23
10004 9.688912e-01
10005 9.383913e-01
...
11432 9.651082e-01
11434 9.603851e-01
11435 9.679081e-01
11691 9.570033e-01
11694 9.700453e-01

[139 rows x 136 columns],
'perf': -0.01974439634331215},
{'model': DecisionTreeClassifier(max_depth=5, min_samples_split=10),
'params_config': {'max_depth': [3, 5, 7],
'min_samples_split': [2, 5, 10],
'min_samples_leaf': [1, 2, 5]},
'nm': 'Single Decision Tree',
'training_feats': ['ttl_night_club',
'ttl_greek_restaurant',
'ttl_mediterranean_restaurant',
'ttl_seafood_restaurant',
'ttl_bar',
'ttl_pizza_restaurant',
'ttl_chinese_restaurant',
'ttl_meal_delivery',
'ttl_bakery',
'ttl_american_restaurant',
'ttl_coffee_shop',
'ttl_dessert_shop',
'EXPENSIVE',
'MODERATE',
'INEXPENSIVE'],
'prediction_var': 'is_gentrified',
'plot': True,
'feature_importances': {'ttl_coffee_shop': 0.3084095798814296,
'ttl_bakery': 0.208503017914835,
'EXPENSIVE': 0.171499524783321,
'INEXPENSIVE': 0.10931077336270391,
'ttl_american_restaurant': 0.08480767401110546,
'ttl_meal_delivery': 0.06188473677461201,
'ttl_dessert_shop': 0.055584693271993064,
'MODERATE': 0.0,
'ttl_chinese_restaurant': 0.0,
'ttl_pizza_restaurant': 0.0,
'ttl_bar': 0.0,
'ttl_seafood_restaurant': 0.0,
'ttl_mediterranean_restaurant': 0.0,
'ttl_greek_restaurant': 0.0,
'ttl_night_club': 0.0},
'train':
    ttl_acai_shop  ttl_afghani_restaurant  ttl_african_restaurant \
10001          0.0              0.0              0.0
10002          0.0              0.0              0.0
10003          1.0              0.0              0.0
10004          0.0              0.0              0.0
10005          0.0              0.0              0.0
...          ...              ...              ...
11432          0.0              0.0              0.0
11434          0.0              0.0              1.0
11435          0.0              0.0              1.0
11691          0.0              0.0              0.0
11694          0.0              0.0              0.0

    ttl_american_restaurant  ttl_amusement_center  ttl_art_gallery \
10001          20.0              0.0              0.0
10002          12.0              0.0              0.0
10003          15.0              0.0              0.0
10004           3.0              0.0              0.0
10005           4.0              0.0              0.0
...          ...              ...              ...
11432           0.0              0.0              0.0
11434           0.0              0.0              0.0
11435           1.0              0.0              0.0
11691           1.0              0.0              0.0
11694           0.0              0.0              0.0

    ttl_asian_grocery_store  ttl_asian_restaurant  ttl_athletic_field \
10001           0.0              10.0              0.0
10002           0.0              10.0              0.0
10003           1.0              12.0              0.0
10004           0.0              0.0              0.0
10005           0.0              0.0              0.0
...          ...              ...              ...
11432           0.0              0.0              0.0
11434           0.0              0.0              0.0

```

```

11435      0.0      0.0      0.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

ttl_auditorium ... ttl_wine_bar ttl_yoga_studio EXPENSIVE \
10001      0.0 ...      8.0      0.0      17.0
10002      0.0 ...      10.0      0.0      12.0
10003      2.0 ...      7.0      1.0      25.0
10004      0.0 ...      1.0      0.0      2.0
10005      0.0 ...      1.0      0.0      4.0
...      ...      ...      ...      ...
11432      0.0 ...      0.0      0.0      0.0
11434      0.0 ...      0.0      0.0      0.0
11435      0.0 ...      0.0      0.0      0.0
11691      0.0 ...      0.0      0.0      0.0
11694      0.0 ...      0.0      0.0      0.0

INEXPENSIVE MODERATE rent rent_change yr is_gentrified \
10001      25.0      90.0 4870.677598 0.319759 2022 False
10002      62.0      75.0 3554.796326 0.287982 2022 False
10003      68.0      108.0 3941.807689 0.345320 2022 True
10004      13.0      11.0 3946.666521 0.227990 2022 False
10005      1.0      5.0 4275.182475 0.199794 2022 False
...      ...      ...      ...      ...
11432      7.0      2.0 2142.269355 0.342445 2022 False
11434      4.0      3.0 2980.772466 0.211454 2022 False
11435      6.0      3.0 1846.797387 0.337604 2022 False
11691      2.0      1.0 2315.852443 0.325885 2022 False
11694      2.0      4.0 3886.183980 0.392739 2022 True

predictions
10001      0.000000
10002      0.000000
10003      1.000000
10004      0.716981
10005      0.310345
...      ...
11432      0.310345
11434      0.310345
11435      0.310345
11691      0.310345
11694      0.310345

[138 rows x 136 columns],
'test':      ttl_acai_shop ttl_afghani_restaurant ttl_african_restaurant \
10001      0.0      0.0      0.0
10002      0.0      0.0      0.0
10003      1.0      0.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...      ...      ...
11432      0.0      0.0      0.0
11434      0.0      0.0      1.0
11435      0.0      0.0      1.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

ttl_american_restaurant ttl_amusement_center ttl_art_gallery \
10001      22.0      0.0      0.0
10002      15.0      0.0      0.0
10003      17.0      0.0      0.0
10004      5.0      0.0      0.0
10005      4.0      0.0      0.0
...      ...      ...
11432      0.0      0.0      0.0
11434      0.0      0.0      0.0
11435      1.0      0.0      0.0
11691      1.0      0.0      0.0
11694      2.0      0.0      0.0

ttl_asian_grocery_store ttl_asian_restaurant ttl_athletic_field \
10001      0.0      12.0      0.0
10002      0.0      16.0      0.0
10003      1.0      22.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...      ...      ...
11432      0.0      1.0      0.0
11434      0.0      0.0      0.0
11435      0.0      0.0      0.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

ttl_auditorium ... ttl_wine_bar ttl_yoga_studio EXPENSIVE \

```

```

10001      0.0 ...      12.0      0.0      18.0
10002      1.0 ...      12.0      0.0      13.0
10003      2.0 ...      9.0      1.0      27.0
10004      0.0 ...      1.0      0.0      3.0
10005      0.0 ...      1.0      0.0      4.0
...      ... ...      ...      ...      ...
11432      0.0 ...      0.0      0.0      0.0
11434      0.0 ...      0.0      0.0      0.0
11435      0.0 ...      0.0      0.0      0.0
11691      0.0 ...      0.0      0.0      0.0
11694      0.0 ...      0.0      0.0      0.0

INEXPENSIVE MODERATE      rent rent_change      yr is_gentrified \
10001      36.0      113.0 4978.021561      0.520362 2023      True
10002      73.0      105.0 3609.439689      0.368481 2023      True
10003      80.0      156.0 4077.081605      0.365517 2023      True
10004      13.0      14.0 4024.260110      0.227990 2023      False
10005      2.0      5.0 4399.192585      0.199794 2023      False
...      ...      ...      ...      ...      ...
11432      12.0      2.0 2309.792636      0.397874 2023      True
11434      5.0      4.0 3135.304935      0.241410 2023      False
11435      7.0      3.0 1954.208071      0.371515 2023      True
11691      3.0      1.0 2271.818549      0.344507 2023      True
11694      2.0      6.0 4072.959038      0.474422 2023      True

predictions
10001      0.000000
10002      1.000000
10003      1.000000
10004      0.716981
10005      0.310345
...      ...
11432      0.310345
11434      0.310345
11435      0.310345
11691      0.310345
11694      0.310345

[139 rows x 136 columns],
'perf': 0.4567037859017066},
{'model': RandomForestClassifier(max_depth=7, n_estimators=50),
 'params_config': {'n_estimators': [50, 100, 200, 300],
 'max_depth': [3, 5, 7],
 'min_samples_split': [2, 5, 10],
 'min_samples_leaf': [1, 2, 5]},
 'nm': 'Random Forests',
 'training_feats': ['ttl_night_club',
 'ttl_greek_restaurant',
 'ttl_mediterranean_restaurant',
 'ttl_seafood_restaurant',
 'ttl_bar',
 'ttl_pizza_restaurant',
 'ttl_chinese_restaurant',
 'ttl_meal_delivery',
 'ttl_bakery',
 'ttl_american_restaurant',
 'ttl_coffee_shop',
 'ttl_dessert_shop',
 'EXPENSIVE',
 'MODERATE',
 'INEXPENSIVE'],
 'prediction_var': 'is_gentrified',
 'plot': True,
 'feature_importances': {'INEXPENSIVE': 0.11843125037276991,
 'ttl_bakery': 0.0986381679265946,
 'MODERATE': 0.09358974317639111,
 'ttl_coffee_shop': 0.07885647052184007,
 'ttl_meal_delivery': 0.07282037509745128,
 'ttl_bar': 0.07267888080624399,
 'EXPENSIVE': 0.0679522810404613,
 'ttl_american_restaurant': 0.06181490392631924,
 'ttl_mediterranean_restaurant': 0.06136420146167838,
 'ttl_chinese_restaurant': 0.0567049804113039,
 'ttl_seafood_restaurant': 0.050512951145514996,
 'ttl_pizza_restaurant': 0.04616568671477388,
 'ttl_dessert_shop': 0.04506534586209119,
 'ttl_night_club': 0.039567918616381405,
 'ttl_greek_restaurant': 0.035836842920184506},
 'train':      ttl_acai_shop      ttl_afghani_restaurant      ttl_african_restaurant \
10001      0.0      0.0      0.0
10002      0.0      0.0      0.0
10003      1.0      0.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0

```

```

...
11432      0.0      0.0      0.0
11434      0.0      0.0      1.0
11435      0.0      0.0      1.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

    ttl_american_restaurant  ttl_amusement_center  ttl_art_gallery  \
10001      20.0      0.0      0.0
10002      12.0      0.0      0.0
10003      15.0      0.0      0.0
10004      3.0      0.0      0.0
10005      4.0      0.0      0.0
...
11432      0.0      0.0      0.0
11434      0.0      0.0      0.0
11435      1.0      0.0      0.0
11691      1.0      0.0      0.0
11694      0.0      0.0      0.0

    ttl_asian_grocery_store  ttl_asian_restaurant  ttl_athletic_field  \
10001      0.0      10.0      0.0
10002      0.0      10.0      0.0
10003      1.0      12.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...
11432      0.0      0.0      0.0
11434      0.0      0.0      0.0
11435      0.0      0.0      0.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

    ttl_auditorium  ...  ttl_wine_bar  ttl_yoga_studio  EXPENSIVE  \
10001      0.0  ...      8.0      0.0      17.0
10002      0.0  ...      10.0      0.0      12.0
10003      2.0  ...      7.0      1.0      25.0
10004      0.0  ...      1.0      0.0      2.0
10005      0.0  ...      1.0      0.0      4.0
...
11432      0.0  ...      0.0      0.0      0.0
11434      0.0  ...      0.0      0.0      0.0
11435      0.0  ...      0.0      0.0      0.0
11691      0.0  ...      0.0      0.0      0.0
11694      0.0  ...      0.0      0.0      0.0

    INEXPENSIVE  MODERATE  rent  rent_change  yr  is_gentrified  \
10001      25.0      90.0  4870.677598  0.319759  2022      False
10002      62.0      75.0  3554.796326  0.287982  2022      False
10003      68.0     108.0  3941.807689  0.345320  2022       True
10004      13.0      11.0  3946.666521  0.227990  2022      False
10005      1.0      5.0   4275.182475  0.199794  2022      False
...
11432      7.0      2.0  2142.269355  0.342445  2022      False
11434      4.0      3.0  2980.772466  0.211454  2022      False
11435      6.0      3.0  1846.797387  0.337604  2022      False
11691      2.0      1.0  2315.852443  0.325885  2022      False
11694      2.0      4.0  3886.183980  0.392739  2022       True

    predictions
10001      0.100000
10002      0.200000
10003      0.600000
10004      0.212696
10005      0.166491
...
11432      0.221068
11434      0.274209
11435      0.220701
11691      0.376228
11694      0.537170

[138 rows x 136 columns],
'test':      ttl_acai_shop  ttl_afghani_restaurant  ttl_african_restaurant  \
10001      0.0      0.0      0.0
10002      0.0      0.0      0.0
10003      1.0      0.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...
11432      0.0      0.0      0.0
11434      0.0      0.0      1.0
11435      0.0      0.0      1.0
11691      0.0      0.0      0.0

```

```

11694      0.0      0.0      0.0

      ttl_american_restaurant ttl_amusement_center ttl_art_gallery \
10001      22.0      0.0      0.0
10002      15.0      0.0      0.0
10003      17.0      0.0      0.0
10004      5.0      0.0      0.0
10005      4.0      0.0      0.0
...      ...      ...      ...
11432      0.0      0.0      0.0
11434      0.0      0.0      0.0
11435      1.0      0.0      0.0
11691      1.0      0.0      0.0
11694      2.0      0.0      0.0

      ttl_asian_grocery_store ttl_asian_restaurant ttl_athletic_field \
10001      0.0      12.0      0.0
10002      0.0      16.0      0.0
10003      1.0      22.0      0.0
10004      0.0      0.0      0.0
10005      0.0      0.0      0.0
...      ...      ...      ...
11432      0.0      1.0      0.0
11434      0.0      0.0      0.0
11435      0.0      0.0      0.0
11691      0.0      0.0      0.0
11694      0.0      0.0      0.0

      ttl_auditorium ... ttl_wine_bar ttl_yoga_studio EXPENSIVE \
10001      0.0 ...      12.0      0.0      18.0
10002      1.0 ...      12.0      0.0      13.0
10003      2.0 ...      9.0      1.0      27.0
10004      0.0 ...      1.0      0.0      3.0
10005      0.0 ...      1.0      0.0      4.0
...      ... ...      ...      ...      ...
11432      0.0 ...      0.0      0.0      0.0
11434      0.0 ...      0.0      0.0      0.0
11435      0.0 ...      0.0      0.0      0.0
11691      0.0 ...      0.0      0.0      0.0
11694      0.0 ...      0.0      0.0      0.0

      INEXPENSIVE MODERATE rent rent_change yr is_gentrified \
10001      36.0      113.0      4978.021561      0.520362      2023      True
10002      73.0      105.0      3609.439689      0.368481      2023      True
10003      80.0      156.0      4077.081605      0.365517      2023      True
10004      13.0      14.0      4024.260110      0.227990      2023      False
10005      2.0      5.0      4399.192585      0.199794      2023      False
...      ...      ...      ...      ...      ...
11432      12.0      2.0      2309.792636      0.397874      2023      True
11434      5.0      4.0      3135.304935      0.241410      2023      False
11435      7.0      3.0      1954.208071      0.371515      2023      True
11691      3.0      1.0      2271.818549      0.344507      2023      True
11694      2.0      6.0      4072.959038      0.474422      2023      True

      predictions
10001      0.280000
10002      0.365000
10003      0.585000
10004      0.349593
10005      0.301267
...      ...
11432      0.336557
11434      0.310991
11435      0.243701
11691      0.295399
11694      0.522254

[139 rows x 136 columns],
'perf': 0.5307895273396422},
{'model': XGBClassifier(base_score=None, booster=None, callbacks=None,
      colsample_bylevel=None, colsample_bynode=None,
      colsample_bytree=None, device=None, early_stopping_rounds=None,
      enable_categorical=False, eval_metric=None, feature_types=None,
      feature_weights=None, gamma=None, grow_policy=None,
      importance_type=None, interaction_constraints=None,
      learning_rate=0.2, max_bin=None, max_cat_threshold=None,
      max_cat_to_onehot=None, max_delta_step=None, max_depth=5,
      max_leaves=None, min_child_weight=None, missing=nan,
      monotone_constraints=None, multi_strategy=None, n_estimators=200,
      n_jobs=None, num_parallel_tree=None, ...),
'params_config': {'n_estimators': [50, 100, 200, 300],
      'max_depth': [3, 5, 7, 9],
      'learning_rate': [0.01, 0.05, 0.1, 0.2]},
'nm': 'XGBoost',

```

```

'training_feats': ['ttl_night_club',
'ttl_greek_restaurant',
'ttl_mediterranean_restaurant',
'ttl_seafood_restaurant',
'ttl_bar',
'ttl_pizza_restaurant',
'ttl_chinese_restaurant',
'ttl_meal_delivery',
'ttl_bakery',
'ttl_american_restaurant',
'ttl_coffee_shop',
'ttl_dessert_shop',
'EXPENSIVE',
'MODERATE',
'INEXPENSIVE'],
'prediction_var': 'is_gentrified',
'plot': True,
'feature_importances': {'EXPENSIVE': 0.14362577,
'ttl_greek_restaurant': 0.10716407,
'ttl_coffee_shop': 0.10611808,
'ttl_bakery': 0.09289351,
'ttl_dessert_shop': 0.072459035,
'ttl_seafood_restaurant': 0.07061396,
'ttl_american_restaurant': 0.06258531,
'MODERATE': 0.05700505,
'ttl_night_club': 0.045059413,
'ttl_chinese_restaurant': 0.043990742,
'ttl_meal_delivery': 0.04184405,
'INEXPENSIVE': 0.041728314,
'ttl_pizza_restaurant': 0.039535638,
'ttl_bar': 0.03785461,
'ttl_mediterranean_restaurant': 0.037522454},
'train':
    ttl_acai_shop    ttl_afghani_restaurant    ttl_african_restaurant \
10001              0.0              0.0              0.0
10002              0.0              0.0              0.0
10003              1.0              0.0              0.0
10004              0.0              0.0              0.0
10005              0.0              0.0              0.0
...              ...              ...
11432              0.0              0.0              0.0
11434              0.0              0.0              1.0
11435              0.0              0.0              1.0
11691              0.0              0.0              0.0
11694              0.0              0.0              0.0

    ttl_american_restaurant    ttl_amusement_center    ttl_art_gallery \
10001              20.0              0.0              0.0
10002              12.0              0.0              0.0
10003              15.0              0.0              0.0
10004              3.0              0.0              0.0
10005              4.0              0.0              0.0
...              ...              ...
11432              0.0              0.0              0.0
11434              0.0              0.0              0.0
11435              1.0              0.0              0.0
11691              1.0              0.0              0.0
11694              0.0              0.0              0.0

    ttl_asian_grocery_store    ttl_asian_restaurant    ttl_athletic_field \
10001              0.0              10.0              0.0
10002              0.0              10.0              0.0
10003              1.0              12.0              0.0
10004              0.0              0.0              0.0
10005              0.0              0.0              0.0
...              ...              ...
11432              0.0              0.0              0.0
11434              0.0              0.0              0.0
11435              0.0              0.0              0.0
11691              0.0              0.0              0.0
11694              0.0              0.0              0.0

    ttl_auditorium    ...    ttl_wine_bar    ttl_yoga_studio    EXPENSIVE \
10001              0.0    ...              8.0              0.0              17.0
10002              0.0    ...              10.0              0.0              12.0
10003              2.0    ...              7.0              1.0              25.0
10004              0.0    ...              1.0              0.0              2.0
10005              0.0    ...              1.0              0.0              4.0
...              ...    ...              ...              ...
11432              0.0    ...              0.0              0.0              0.0
11434              0.0    ...              0.0              0.0              0.0
11435              0.0    ...              0.0              0.0              0.0
11691              0.0    ...              0.0              0.0              0.0
11694              0.0    ...              0.0              0.0              0.0

```

	INEXPENSIVE	MODERATE	rent	rent_change	yr	is_gentrified	\
10001	25.0	90.0	4870.677598	0.319759	2022	False	
10002	62.0	75.0	3554.796326	0.287982	2022	False	
10003	68.0	108.0	3941.807689	0.345320	2022	True	
10004	13.0	11.0	3946.666521	0.227990	2022	False	
10005	1.0	5.0	4275.182475	0.199794	2022	False	
...	...	...	...	...	...	...	...
11432	7.0	2.0	2142.269355	0.342445	2022	False	
11434	4.0	3.0	2980.772466	0.211454	2022	False	
11435	6.0	3.0	1846.797387	0.337604	2022	False	
11691	2.0	1.0	2315.852443	0.325885	2022	False	
11694	2.0	4.0	3886.183980	0.392739	2022	True	

predictions

10001	0.005563
10002	0.141376
10003	0.783748
10004	0.072654
10005	0.031763
...	...
11432	0.012161
11434	0.054056
11435	0.017581
11691	0.043912
11694	0.938231

[138 rows x 136 columns],				
'test':	t1l_acai_shop	t1l_afghani_restaurant	t1l_african_restaurant	\
10001	0.0	0.0	0.0	
10002	0.0	0.0	0.0	
10003	1.0	0.0	0.0	
10004	0.0	0.0	0.0	
10005	0.0	0.0	0.0	
...	...	...	...	
11432	0.0	0.0	0.0	
11434	0.0	0.0	1.0	
11435	0.0	0.0	1.0	
11691	0.0	0.0	0.0	
11694	0.0	0.0	0.0	

	t1l_american_restaurant	t1l_amusement_center	t1l_art_gallery	\
10001	22.0	0.0	0.0	
10002	15.0	0.0	0.0	
10003	17.0	0.0	0.0	
10004	5.0	0.0	0.0	
10005	4.0	0.0	0.0	
...	...	...	...	
11432	0.0	0.0	0.0	
11434	0.0	0.0	0.0	
11435	1.0	0.0	0.0	
11691	1.0	0.0	0.0	
11694	2.0	0.0	0.0	

	t1l_asian_grocery_store	t1l_asian_restaurant	t1l_athletic_field	\
10001	0.0	12.0	0.0	
10002	0.0	16.0	0.0	
10003	1.0	22.0	0.0	
10004	0.0	0.0	0.0	
10005	0.0	0.0	0.0	
...	...	...	...	
11432	0.0	1.0	0.0	
11434	0.0	0.0	0.0	
11435	0.0	0.0	0.0	
11691	0.0	0.0	0.0	
11694	0.0	0.0	0.0	

	t1l_auditorium	...	t1l_wine_bar	t1l_yoga_studio	EXPENSIVE	\
10001	0.0	...	12.0	0.0	18.0	
10002	1.0	...	12.0	0.0	13.0	
10003	2.0	...	9.0	1.0	27.0	
10004	0.0	...	1.0	0.0	3.0	
10005	0.0	...	1.0	0.0	4.0	
...	...	...	...	...	...	
11432	0.0	...	0.0	0.0	0.0	
11434	0.0	...	0.0	0.0	0.0	
11435	0.0	...	0.0	0.0	0.0	
11691	0.0	...	0.0	0.0	0.0	
11694	0.0	...	0.0	0.0	0.0	

	INEXPENSIVE	MODERATE	rent	rent_change	yr	is_gentrified	\
10001	36.0	113.0	4978.021561	0.520362	2023	True	
10002	73.0	105.0	3609.439689	0.368481	2023	True	
10003	80.0	156.0	4077.081605	0.365517	2023	True	
10004	13.0	14.0	4024.260110	0.227990	2023	False	

```

10005      2.0      5.0  4399.192585      0.199794  2023      False
...      ...      ...      ...      ...      ...
11432     12.0      2.0  2309.792636      0.397874  2023      True
11434      5.0      4.0  3135.304935      0.241410  2023      False
11435      7.0      3.0  1954.208071      0.371515  2023      True
11691      3.0      1.0  2271.818549      0.344507  2023      True
11694      2.0      6.0  4072.959038      0.474422  2023      True

```

```

      predictions
10001      0.292490
10002      0.401822
10003      0.726041
10004      0.168867
10005      0.062209
...      ...
11432      0.006215
11434      0.430327
11435      0.015427
11691      0.092406
11694      0.812118

```

```

[139 rows x 136 columns],
'perf': 0.5190943638301379}]

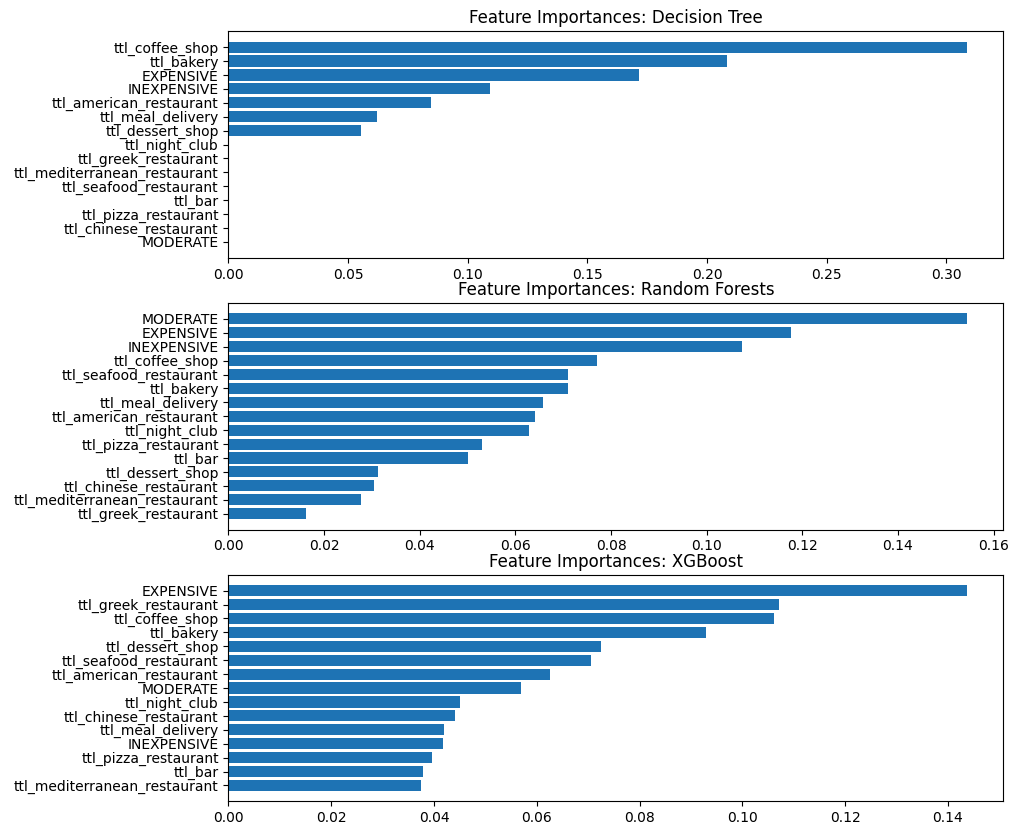
```

In []: *#### output feature importances*

```

feature_importances = pd.DataFrame([business_models[i]['feature_importances'] for i in range(1,4)]).T
feature_importances.columns = ['Decision Tree', 'Random Forests', 'XGBoost']
fig,ax=plt.subplots(3,1,figsize=(10,10))
for idx,f in enumerate(feature_importances.columns):
    v=feature_importances[[f]].sort_values(by=[f])
    ax[idx].barh(
        v.index,
        v[f],
    )
    ax[idx].set_title(f'Feature Importances: {f}')
plt.show()

```



```
In [ ]: #plot the spatial results...

geo_zip_nyc = gpd.GeoDataFrame(pd.read_csv('datasets/nyc_open_datasets/Modified_Zip_Code_Tabulation_Areas_MODZCTA_20250421.csv'))
geo_zip_nyc['zip'] = geo_zip_nyc['MODZCTA'].astype(str)

geo_zip_nyc = geo_zip_nyc\
    .rename(columns={
        'the_geom': 'geometry'
    })\
    .set_index('zip')[['geometry']]

geo_zip_nyc['geometry'] = geo_zip_nyc['geometry'].apply(loads)
xgb_r = business_models[-1]

model_scores = xgb_r['test'][['predictions']].rename(
    columns={
        'predictions': 'Gentrification Hotspot Score'
    }
)

w_geo = gpd.GeoDataFrame(pd.concat((geo_zip_nyc,model_scores),axis=1).dropna(), crs="EPSG:4326")

fig = px.choropleth_mapbox(
    w_geo,
    geojson=w_geo['geometry'],
    locations=w_geo.index,
    color="Gentrification Hotspot Score",
    featureidkey="id",
    center={"lat": w_geo.geometry.centroid.y.mean(), "lon": w_geo.geometry.centroid.x.mean()},
    zoom=10,
    mapbox_style="carto-positron",
)
```

```
fig.update_layout(margin={"r": 0, "t": 0, "l": 0, "b": 0})
fig.show()
```

Socio-demographic sub-model

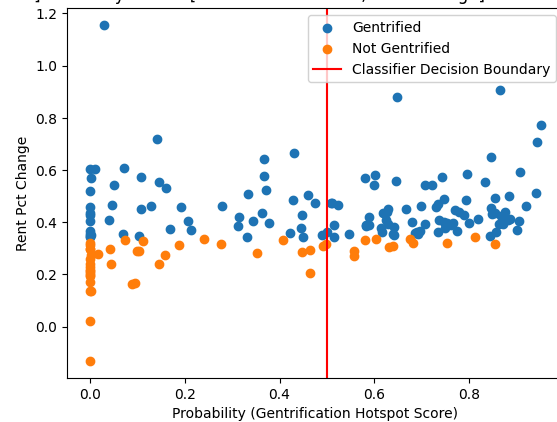
```
In [ ]: feats = [ #use insights per Andrew's result...
    'less_than_9th_grade',
    '9th_to_12th_grade',
    'high_school_graduate',
    'some_college_no_degree',
    'associates_degree',
    'bachelors_degree',
    'graduate_or_professional_degree',
    'pct_children_teens_5-19',
    'pct_young_adults_20-34',
    'pct_middle_aged_35-59',
    'pct_older_adults_60+',
    'pct_1_person',
    'pct_2_person',
    'pct_3_person',
    'pct_4_or_more_person',
    'median_rent'
]

acs_2015_tr = acs_2015_tr.rename(
    columns={
        'median_rent_change': 'rent_change'
    }
)

acs_2015_tr['is_gentrified'] = acs_2015_tr['rent_change'] > 0.343

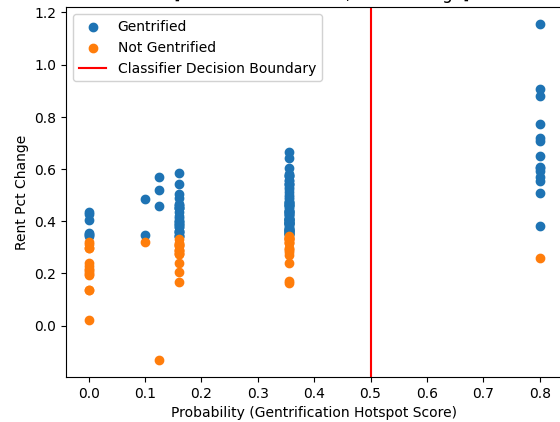
training_feats = feats[:-1]
train = acs_2015_tr.loc[acs_2015_tr['year'].astype(int).between(2021,2022)]
test = acs_2015_tr.loc[acs_2015_tr['year'].astype(int)==2023]
acs_models = run_all_models(train, test, training_feats, target='is_gentrified')
acs_models
```

[BASELINE] NaiveBayes: Corr[Gentrification Score, Rent Change]=0.2976761914369301



```
*** [BASELINE] NaiveBayes Accuracy: 0.651685393258427 ***
*** [BASELINE] NaiveBayes F1: 0.7129629629629629 ***
*** Feature Importance Scores ***
{}
```

Single Decision Tree: Corr[Gentrification Score, Rent Change]=0.5950758733968496



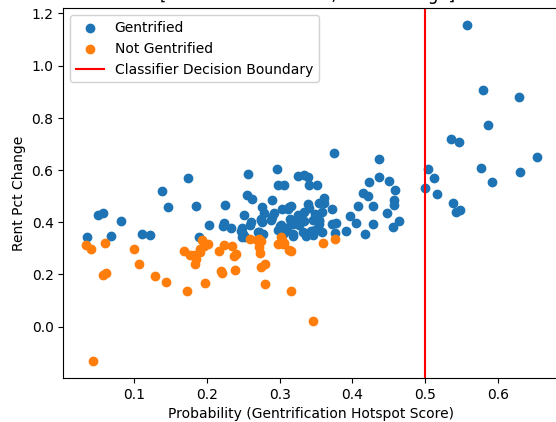
*** Single Decision Tree Accuracy: 0.3539325842696629 ***

*** Single Decision Tree F1: 0.1958041958041958 ***

*** Feature Importance Scores ***

{'pct_older_adults_60+': 0.3373625017015004, 'less_than_9th_grade': 0.28951092489347763, 'pct_4_or_more_person': 0.24163447485588077, 'associates_degree': 0.125071114417129, 'some_college_no_degree': 0.0064209841320122096, 'pct_3_person': 0.0, 'pct_2_person': 0.0, 'pct_1_person': 0.0, 'pct_middle_aged_35-59': 0.0, 'pct_young_adults_20-34': 0.0, 'pct_children_teens_5-19': 0.0, 'graduate_or_professional_degree': 0.0, 'bachelors_degree': 0.0, 'high_school_graduate': 0.0, '9th_to_12th_grade': 0.0}

Random Forests: Corr[Gentrification Score, Rent Change]=0.628994943978679



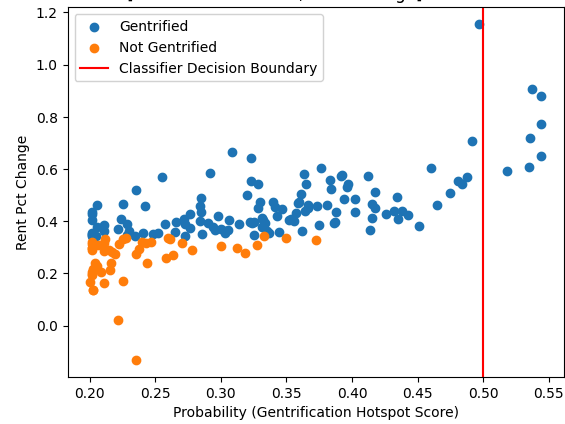
*** Random Forests Accuracy: 0.37640449438202245 ***

*** Random Forests F1: 0.23448275862068965 ***

*** Feature Importance Scores ***

{'pct_older_adults_60+': 0.1219865590900348, 'associates_degree': 0.11541571486531832, 'pct_1_person': 0.09952773684403096, 'pct_young_adults_20-34': 0.08605738919884366, 'less_than_9th_grade': 0.07418721856132897, 'pct_4_or_more_person': 0.06267369089919754, 'pct_3_person': 0.06176123561621981, 'bachelors_degree': 0.059048976017301705, 'graduate_or_professional_degree': 0.05606148944742956, 'pct_2_person': 0.053278909471304634, 'pct_middle_aged_35-59': 0.05096192770789523, '9th_to_12th_grade': 0.04919584779641835, 'high_school_graduate': 0.039311929692036465, 'some_college_no_degree': 0.03791632664153296, 'pct_children_teens_5-19': 0.03261504815110721}

XGBoost: Corr[Gentrification Score, Rent Change]=0.7112388132701446



*** XGBoost Accuracy: 0.3202247191011236 ***

*** XGBoost F1: 0.1037037037037037 ***

*** Feature Importance Scores ***

{'less_than_9th_grade': 0.16573013, 'pct_older_adults_60+': 0.11107012, 'pct_4_or_more_person': 0.08674421, 'pct_1_person': 0.071845375, 'high_school_graduate': 0.067065425, 'pct_middle_aged_35-59': 0.06407279, 'pct_3_person': 0.0635566, 'graduate_or_professional_degree': 0.06228263, 'associates_degree': 0.060143106, 'bachelors_degree': 0.058995467, '9th_to_12th_grade': 0.055510644, 'pct_children_teens_5-19': 0.042493638, 'pct_young_adults_20-34': 0.040093504, 'pct_2_person': 0.0320735, 'some_college_no_degree': 0.018322786}

```

Out[ ]: [{'model': GaussianNB(),
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'nm': '[BASELINE] NaiveBayes',
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'some_college_no_degree',
'associates_degree',
'bachelors_degree',
'graduate_or_professional_degree',
'pct_children_teens_5-19',
'pct_young_adults_20-34',
'pct_middle_aged_35-59',
'pct_older_adults_60+',
'pct_1_person',
'pct_2_person',
'pct_3_person',
'pct_4_or_more_person'],
'prediction_var': 'is_gentrified',
'plot': True,
'feature_importances': {},
'train':
zip_code year less_than_9th_grade 9th_to_12th_grade \
212 10001 2022 3.494788 5.607697
213 10002 2022 16.173382 13.851883
214 10003 2022 2.698622 1.544766
215 10004 2022 0.446180 0.446180
216 10005 2022 0.935399 1.520023
.. ...
631 11691 2021 10.376808 9.673652
632 11692 2021 9.720558 7.910175
633 11693 2021 8.382187 8.443896
634 11694 2021 4.306104 6.158518
635 11697 2021 0.000000 4.119986

high_school_graduate some_college_no_degree associates_degree \
212 8.498797 8.352592 3.117483
213 18.186442 9.840530 3.850571
214 5.348718 5.990349 2.458685
215 0.864473 3.457892 0.306748
216 4.998538 2.850044 0.950015
.. ...
631 29.663373 16.167916 7.888897
632 29.155365 18.677853 8.219264
633 25.619665 13.966883 9.143269
634 23.783784 15.663529 5.460067
635 14.564510 19.804843 10.336104

bachelors_degree graduate_or_professional_degree \
212 42.027072 28.901571
213 26.392899 11.704292
214 43.849244 38.109616
215 44.283324 50.195204
216 56.562409 32.183572
.. ...
631 14.266358 11.962997
632 16.003280 10.313505
633 19.685282 14.758819
634 23.467962 21.160036
635 26.816046 24.358511

pct_children_teens_5-19 ... pct_middle_aged_35-59_change \
212 10.224411 ... 0.098539
213 11.626023 ... -0.074749
214 16.704716 ... -0.096026
215 9.849312 ... 0.204775
216 5.522100 ... -0.035605
.. ...
631 24.563188 ... -0.118124
632 21.766305 ... -0.038763
633 19.862868 ... -0.223210
634 17.229850 ... 0.027667
635 14.754098 ... -0.313308

pct_older_adults_60+_change pct_1_person_change pct_2_person_change \
212 -0.099233 0.073848 -0.049592
213 0.227172 0.103821 0.096883
214 0.130662 -0.054713 0.024810
215 -0.126447 -0.252555 0.209940
216 0.746014 0.112869 -0.135661
.. ...
631 0.124117 0.041773 -0.045256
632 0.105594 0.257573 0.008721
633 0.270018 -0.215785 0.264911
634 0.084566 -0.018356 0.082177

```

```

635          0.337800          0.039733          -0.100402

      pct_3_person_change  pct_4_or_more_person_change \
212          -0.137526          -0.160102
213          -0.124331          -0.346550
214           0.243424           0.114622
215           1.010748          -0.156068
216          -0.218806           0.529741
..          ...
631           0.184168          -0.090151
632          -0.235028          -0.037315
633          -0.079417           0.093299
634          -0.094960          -0.011590
635           0.156551           0.008477

      household_median_income_lyr_change  rent_change  is_gentrified \
212           0.227048           0.319759          False
213           0.285714           0.287982          False
214           0.557427           0.345320           True
215           0.942861           0.227990          False
216           0.528947           0.199794          False
..          ...
631           0.305880           0.255121          False
632           0.269922           0.495868           True
633           0.214091           0.112299          False
634           0.180427           0.284653          False
635           0.211222          -1.000001          False

      predictions
212  3.015947e-04
213  1.547286e-01
214  9.410081e-05
215  4.065071e-05
216  8.937296e-09
..    ...
631  6.532617e-01
632  7.591454e-01
633  7.219725e-01
634  5.256497e-01
635  6.466562e-04

[359 rows x 38 columns],
'test':   zip_code  year  less_than_9th_grade  9th_to_12th_grade \
0      10001  2023         2.112304         5.034325
1      10002  2023        13.788751        15.215928
2      10003  2023         2.982706         1.313485
3      10004  2023         0.531561         0.365449
4      10005  2023         0.887372         1.665529
..    ...
205    11436  2023         6.311237         9.230353
207    11691  2023        11.703595        10.241795
208    11692  2023         6.898884         7.379100
209    11693  2023         7.781227         8.177245
210    11694  2023         4.847583         4.899359

      high_school_graduate  some_college_no_degree  associates_degree \
0           8.638444           9.514170           3.199261
1          17.723585          10.347033           3.970177
2           4.802430           6.170644           2.531195
3           1.661130           6.378738           0.166113
4           5.870307           6.484642           0.832765
..    ...
205          36.083519          17.974187           8.169471
207          27.801784          18.210444           8.761607
208          31.268177          18.369970           9.543456
209          27.689179          14.224553           6.379107
210          25.124587          16.652644           5.210019

      bachelors_degree  graduate_or_professional_degree \
0          41.471572           30.029924
1          26.542462           12.412064
2          44.729641           37.469899
3          42.624585           48.272425
4          50.552901           33.706485
..    ...
205          15.048314           7.182918
207          12.567804           10.712972
208          16.307068           10.233345
209          21.920154           13.828535
210          22.516342           20.749466

      pct_children_teens_5-19  ...  pct_middle_aged_35-59_change \
0          10.653736  ...           0.088749
1          11.974787  ...          -0.077519

```

```

2          17.103576 ...          -0.126590
3          11.561290 ...          0.145315
4           5.877896 ...          0.024834
..          ... ..
205         18.910316 ...          -0.077443
207         22.684269 ...          -0.093891
208         24.222480 ...          -0.106917
209         18.421858 ...          -0.293290
210         18.987401 ...          0.019953

    pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change  \
0          -0.058638          0.026323          -0.000095
1           0.193148          0.089236          0.118224
2           0.142495          -0.052623          0.010979
3          -0.156268          -0.194850          0.159015
4           2.030959          0.134394          -0.130185
..          ... ..
205         0.294014          0.081634          0.046443
207         0.248278          0.115528          -0.019349
208         0.198880          0.320081          -0.088869
209         0.496218          -0.100137          0.237027
210         0.068398          -0.051488          0.043743

    pct_3_person_change  pct_4_or_more_person_change  \
0          -0.075879          -0.092145
1          -0.130464          -0.343500
2           0.398857          -0.054939
3           0.690141          -0.057121
4          -0.426038          0.827429
..          ... ..
205         -0.028139          -0.046383
207          0.071431          -0.117662
208         -0.277919          0.049429
209         -0.174290          -0.004850
210          0.065194          -0.027958

    household_median_income_lyr_change  rent_change  is_gentrified  \
0          0.421562          0.520362          True
1          0.379499          0.368481          True
2          0.566464          0.365517          True
3          0.843012          0.227990          False
4          0.705477          0.199794          False
..          ... ..
205         0.345270          0.464085          True
207         0.357043          0.344507          True
208         0.175185          0.504132          True
209         0.273043          0.281283          False
210         0.382161          0.474422          True

    predictions
0    1.062441e-03
1    2.127889e-01
2    1.284240e-04
3    2.370070e-04
4    9.468242e-08
..    ...
205  2.785053e-01
207  5.138396e-01
208  4.604790e-01
209  3.527884e-01
210  4.737951e-01

[178 rows x 38 columns],
'perf': 0.2976761914369301},
{'model': DecisionTreeClassifier(max_depth=3, min_samples_leaf=5),
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'min_samples_split': [2, 5, 10],
'min_samples_leaf': [1, 2, 5]},
'nm': 'Single Decision Tree',
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'high_school_graduate',
'some_college_no_degree',
'associates_degree',
'bachelors_degree',
'graduate_or_professional_degree',
'pct_children_teens_5-19',
'pct_young_adults_20-34',
'pct_middle_aged_35-59',
'pct_older_adults_60+',
'pct_1_person',
'pct_2_person',
'pct_3_person',
'pct_4_or_more_person'],

```

```

'prediction_var': 'is_gentrified',
'plot': True,
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'less_than_9th_grade': 0.28951092489347763,
'pct_4_or_more_person': 0.24163447485588077,
'associates_degree': 0.125071114417129,
'some_college_no_degree': 0.0064209841320122096,
'pct_3_person': 0.0,
'pct_2_person': 0.0,
'pct_1_person': 0.0,
'pct_middle_aged_35-59': 0.0,
'pct_young_adults_20-34': 0.0,
'pct_children_teens_5-19': 0.0,
'graduate_or_professional_degree': 0.0,
'bachelors_degree': 0.0,
'high_school_graduate': 0.0,
'9th_to_12th_grade': 0.0},
'train':
  zip_code  year  less_than_9th_grade  9th_to_12th_grade  \
212  10001  2022      3.494788      5.607697
213  10002  2022      16.173382     13.851883
214  10003  2022      2.698622      1.544766
215  10004  2022      0.446180      0.446180
216  10005  2022      0.935399      1.520023
..      ..      ..
631  11691  2021     10.376808      9.673652
632  11692  2021      9.720558      7.910175
633  11693  2021      8.382187      8.443896
634  11694  2021      4.306104      6.158518
635  11697  2021      0.000000      4.119986

  high_school_graduate  some_college_no_degree  associates_degree  \
212      8.498797      8.352592      3.117483
213     18.186442      9.840530      3.850571
214      5.348718      5.990349      2.458685
215      0.864473      3.457892      0.306748
216      4.998538      2.850044      0.950015
..      ..      ..
631     29.663373     16.167916      7.888897
632     29.155365     18.677853      8.219264
633     25.619665     13.966883      9.143269
634     23.783784     15.663529      5.460067
635     14.564510     19.804843     10.336104

  bachelors_degree  graduate_or_professional_degree  \
212     42.027072      28.901571
213     26.392899      11.704292
214     43.849244      38.109616
215     44.283324      50.195204
216     56.562409      32.183572
..      ..      ..
631     14.266358      11.962997
632     16.003280      10.313505
633     19.685282      14.758819
634     23.467962      21.160036
635     26.816046      24.358511

  pct_children_teens_5-19  ...  pct_middle_aged_35-59_change  \
212      10.224411  ...      0.098539
213      11.626023  ...     -0.074749
214      16.704716  ...     -0.096026
215       9.849312  ...      0.204775
216      5.522100  ...     -0.035605
..      ..      ..
631      24.563188  ...     -0.118124
632      21.766305  ...     -0.038763
633      19.862868  ...     -0.223210
634      17.229850  ...      0.027667
635      14.754098  ...     -0.313308

  pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change  \
212      -0.099233      0.073848      -0.049592
213       0.227172      0.103821      0.096883
214       0.130662     -0.054713      0.024810
215      -0.126447     -0.252555      0.209940
216       0.746014      0.112869     -0.135661
..      ..      ..
631       0.124117      0.041773     -0.045256
632       0.105594      0.257573      0.008721
633       0.270018     -0.215785      0.264911
634       0.084566     -0.018356      0.082177
635       0.337800      0.039733     -0.100402

  pct_3_person_change  pct_4_or_more_person_change  \
212      -0.137526      -0.160102

```

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213      -0.124331      -0.346550
214      0.243424      0.114622
215      1.010748      -0.156068
216      -0.218806      0.529741
..      ...
631      0.184168      -0.090151
632      -0.235028      -0.037315
633      -0.079417      0.093299
634      -0.094960      -0.011590
635      0.156551      0.008477

household_median_income_1yr_change  rent_change  is_gentrified \
212      0.227048      0.319759      False
213      0.285714      0.287982      False
214      0.557427      0.345320      True
215      0.942861      0.227990      False
216      0.528947      0.199794      False
..      ...
631      0.305880      0.255121      False
632      0.269922      0.495868      True
633      0.214091      0.112299      False
634      0.180427      0.284653      False
635      0.211222      -1.000001      False

predictions
212      0.354839
213      0.354839
214      0.354839
215      0.000000
216      0.000000
..      ...
631      0.354839
632      0.354839
633      0.160000
634      0.354839
635      0.125000

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1      10002  2023      13.788751      15.215928
2      10003  2023      2.982706      1.313405
3      10004  2023      0.531561      0.365449
4      10005  2023      0.887372      1.665529
..      ...
205     11436  2023      6.311237      9.230353
207     11691  2023      11.703595      10.241795
208     11692  2023      6.898884      7.379100
209     11693  2023      7.781227      8.177245
210     11694  2023      4.847583      4.899359

high_school_graduate  some_college_no_degree  associates_degree \
0      8.638444      9.514170      3.199261
1      17.723585      10.347033      3.970177
2      4.802430      6.170644      2.531195
3      1.661130      6.378738      0.166113
4      5.870307      6.484642      0.832765
..      ...
205     36.083519      17.974187      8.169471
207     27.801784      18.210444      8.761607
208     31.268177      18.369970      9.543456
209     27.689179      14.224553      6.379107
210     25.124587      16.652644      5.210019

bachelors_degree  graduate_or_professional_degree \
0      41.471572      30.029924
1      26.542462      12.412064
2      44.729641      37.469899
3      42.624585      48.272425
4      50.552901      33.706485
..      ...
205     15.048314      7.182918
207     12.567804      10.712972
208     16.307068      10.233345
209     21.920154      13.828535
210     22.516342      20.749466

pct_children_teens_5-19  ...  pct_middle_aged_35-59_change \
0      10.653736      ...      0.088749
1      11.974787      ...      -0.077519
2      17.103576      ...      -0.126590
3      11.561290      ...      0.145315
4      5.877896      ...      0.024834
..      ...

```

```

205      18.910316 ...      -0.077443
207      22.684269 ...      -0.093891
208      24.222480 ...      -0.106917
209      18.421858 ...      -0.293290
210      18.987401 ...      0.019953

      pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change \
0      -0.058638      0.026323      -0.000095
1      0.193148      0.089236      0.118224
2      0.142495      -0.052623      0.010979
3      -0.156268      -0.194850      0.159015
4      2.030959      0.134394      -0.130185
..      ...      ...      ...
205      0.294014      0.081634      0.046443
207      0.248278      0.115528      -0.019349
208      0.198880      0.320081      -0.088869
209      0.496218      -0.100137      0.237027
210      0.068398      -0.051488      0.043743

      pct_3_person_change  pct_4_or_more_person_change \
0      -0.075879      -0.092145
1      -0.130464      -0.343500
2      0.398857      -0.054939
3      0.690141      -0.057121
4      -0.426038      0.827429
..      ...      ...
205      -0.028139      -0.046383
207      0.071431      -0.117662
208      -0.277919      0.049429
209      -0.174290      -0.004850
210      0.065194      -0.027958

      household_median_income_lyr_change  rent_change  is_gentrified \
0      0.421562      0.520362      True
1      0.379499      0.368481      True
2      0.566464      0.365517      True
3      0.843012      0.227990      False
4      0.705477      0.199794      False
..      ...      ...
205      0.345270      0.464085      True
207      0.357043      0.344507      True
208      0.175185      0.504132      True
209      0.273043      0.281283      False
210      0.382161      0.474422      True

      predictions
0      0.125000
1      0.354839
2      0.354839
3      0.000000
4      0.000000
..      ...
205      0.354839
207      0.160000
208      0.160000
209      0.354839
210      0.354839

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'bachelors_degree',
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'pct_middle_aged_35-59',
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'less_than_9th_grade': 0.07418721856132897,
'pct_4_or_more_person': 0.06267369089919754,
'pct_3_person': 0.06176123561621981,
'bachelors_degree': 0.059048976017301705,
'graduate_or_professional_degree': 0.05606148944742956,
'pct_2_person': 0.053278909471304634,
'pct_middle_aged_35-59': 0.05096192770789523,
'9th_to_12th_grade': 0.04919584779641835,
'high_school_graduate': 0.039311929692036465,
'some_college_no_degree': 0.03791632664153296,
'pct_children_teens_5-19': 0.03261504815110721},
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212 10001 2022 3.494788 5.607697
213 10002 2022 16.173382 13.851883
214 10003 2022 2.698622 1.544766
215 10004 2022 0.446180 0.446180
216 10005 2022 0.935399 1.520023
.. ...
631 11691 2021 10.376808 9.673652
632 11692 2021 9.720558 7.910175
633 11693 2021 8.382187 8.443896
634 11694 2021 4.306104 6.158518
635 11697 2021 0.000000 4.119986

high_school_graduate some_college_no_degree associates_degree \
212 8.498797 8.352592 3.117483
213 18.186442 9.840530 3.850571
214 5.348718 5.990349 2.458685
215 0.864473 3.457892 0.306748
216 4.998538 2.850044 0.950015
.. ...
631 29.663373 16.167916 7.888897
632 29.155365 18.677853 8.219264
633 25.619665 13.966883 9.143269
634 23.783784 15.663529 5.460067
635 14.564510 19.804843 10.336104

bachelors_degree graduate_or_professional_degree \
212 42.027072 28.901571
213 26.392899 11.704292
214 43.849244 38.109616
215 44.283324 50.195204
216 56.562409 32.183572
.. ...
631 14.266358 11.962997
632 16.003280 10.313505
633 19.685282 14.758819
634 23.467962 21.160036
635 26.816046 24.358511

pct_children_teens_5-19 ... pct_middle_aged_35-59_change \
212 10.224411 ... 0.098539
213 11.626023 ... -0.074749
214 16.704716 ... -0.096026
215 9.849312 ... 0.204775
216 5.522100 ... -0.035605
.. ...
631 24.563188 ... -0.118124
632 21.766305 ... -0.038763
633 19.862868 ... -0.223210
634 17.229850 ... 0.027667
635 14.754098 ... -0.313308

pct_older_adults_60+_change pct_1_person_change pct_2_person_change \
212 -0.099233 0.073848 -0.049592
213 0.227172 0.103821 0.096883
214 0.130662 -0.054713 0.024810
215 -0.126447 -0.252555 0.209940
216 0.746014 0.112869 -0.135661
.. ...
631 0.124117 0.041773 -0.045256
632 0.105594 0.257573 0.008721
633 0.270018 -0.215785 0.264911
634 0.084566 -0.018356 0.082177
635 0.337800 0.039733 -0.100402

pct_3_person_change pct_4_or_more_person_change \
212 -0.137526 -0.160102
213 -0.124331 -0.346550
214 0.243424 0.114622
215 1.010748 -0.156068

```

```

216      -0.218806      0.529741
..      ...
631      0.184168      -0.090151
632      -0.235028      -0.037315
633      -0.079417      0.093299
634      -0.094960      -0.011590
635      0.156551      0.008477

household_median_income_1yr_change  rent_change  is_gentrified \
212      0.227048      0.319759      False
213      0.285714      0.287982      False
214      0.557427      0.345320      True
215      0.942861      0.227990      False
216      0.528947      0.199794      False
..      ...
631      0.305880      0.255121      False
632      0.269922      0.495868      True
633      0.214091      0.112299      False
634      0.180427      0.284653      False
635      0.211222      -1.000001      False

predictions
212      0.179741
213      0.260086
214      0.290673
215      0.225070
216      0.070103
..      ...
631      0.287355
632      0.347040
633      0.217242
634      0.343556
635      0.200884

[359 rows x 38 columns],
'test':   zip_code  year  less_than_9th_grade  9th_to_12th_grade \
0      10001  2023      2.112304      5.034325
1      10002  2023      13.788751      15.215928
2      10003  2023      2.982706      1.313485
3      10004  2023      0.531561      0.365449
4      10005  2023      0.887372      1.665529
..      ...
205     11436  2023      6.311237      9.230353
207     11691  2023      11.703595      10.241795
208     11692  2023      6.898884      7.379100
209     11693  2023      7.781227      8.177245
210     11694  2023      4.847583      4.899359

high_school_graduate  some_college_no_degree  associates_degree \
0      8.638444      9.514170      3.199261
1      17.723585      10.347033      3.970177
2      4.802430      6.170644      2.531195
3      1.661130      6.378738      0.166113
4      5.870307      6.484642      0.832765
..      ...
205     36.083519      17.974187      8.169471
207     27.801784      18.210444      8.761607
208     31.268177      18.369970      9.543456
209     27.689179      14.224553      6.379107
210     25.124587      16.652644      5.210019

bachelors_degree  graduate_or_professional_degree \
0      41.471572      30.029924
1      26.542462      12.412064
2      44.729641      37.469899
3      42.624585      48.272425
4      50.552901      33.706485
..      ...
205     15.048314      7.182918
207     12.567804      10.712972
208     16.307068      10.233345
209     21.920154      13.828535
210     22.516342      20.749466

pct_children_teens_5-19  ...  pct_middle_aged_35-59_change \
0      10.653736      ...      0.088749
1      11.974787      ...      -0.077519
2      17.103576      ...      -0.126590
3      11.561290      ...      0.145315
4      5.877896      ...      0.024834
..      ...
205     18.910316      ...      -0.077443
207     22.684269      ...      -0.093891
208     24.222480      ...      -0.106917

```

```

209      18.421858 ...      -0.293290
210      18.987401 ...      0.019953

      pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change  \
0      -0.058638      0.026323      -0.000095
1      0.193148      0.089236      0.118224
2      0.142495      -0.052623      0.010979
3      -0.156268      -0.194850      0.159015
4      2.030959      0.134394      -0.130185
..      ...
205      0.294014      0.081634      0.046443
207      0.248278      0.115528      -0.019349
208      0.198880      0.320081      -0.088869
209      0.496218      -0.100137      0.237027
210      0.068398      -0.051488      0.043743

      pct_3_person_change  pct_4_or_more_person_change  \
0      -0.075879      -0.092145
1      -0.130464      -0.343500
2      0.398857      -0.054939
3      0.690141      -0.057121
4      -0.426038      0.827429
..      ...
205      -0.028139      -0.046383
207      0.071431      -0.117662
208      -0.277919      0.049429
209      -0.174290      -0.004850
210      0.065194      -0.027958

      household_median_income_1yr_change  rent_change  is_gentrified  \
0      0.421562      0.520362      True
1      0.379499      0.368481      True
2      0.566464      0.365517      True
3      0.843012      0.227990      False
4      0.705477      0.199794      False
..      ...
205      0.345270      0.464085      True
207      0.357043      0.344507      True
208      0.175185      0.504132      True
209      0.273043      0.281283      False
210      0.382161      0.474422      True

      predictions
0      0.138812
1      0.297990
2      0.246275
3      0.273989
4      0.056744
..      ...
205      0.316184
207      0.188999
208      0.255469
209      0.273155
210      0.360557

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'pct_2_person': 0.0320735,
'some_college_no_degree': 0.018322786},
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212  10001  2022      3.494788      5.607697
213  10002  2022     16.173382     13.851883
214  10003  2022      2.698622      1.544766
215  10004  2022      0.446180      0.446180
216  10005  2022      0.935399      1.520023
..      ...      ...
631  11691  2021     10.376808      9.673652
632  11692  2021      9.720558      7.910175
633  11693  2021      8.382187      8.443896
634  11694  2021      4.306104      6.158518
635  11697  2021      0.000000      4.119986

  high_school_graduate  some_college_no_degree  associates_degree \
212      8.498797      8.352592      3.117483
213     18.186442      9.840530      3.850571
214      5.348718      5.990349      2.458685
215      0.864473      3.457892      0.306748
216      4.998538      2.850044      0.950015
..      ...
631     29.663373     16.167916      7.888897
632     29.155365     18.677853      8.219264
633     25.619665     13.966883      9.143269
634     23.783784     15.663529      5.460067
635     14.564510     19.804843     10.336104

  bachelors_degree  graduate_or_professional_degree \
212      42.027072      28.901571
213      26.392899      11.704292
214      43.849244      38.109616
215      44.283324      50.195204
216      56.562409      32.183572
..      ...
631     14.266358      11.962997
632     16.003280      10.313505
633     19.685282      14.758819
634     23.467962      21.160036
635     26.816046      24.358511

  pct_children_teens_5-19  ...  pct_middle_aged_35-59_change \
212      10.224411  ...      0.098539
213      11.626023  ...     -0.074749
214      16.704716  ...     -0.096026
215       9.849312  ...      0.204775
216      5.522100  ...     -0.035605
..      ...
631      24.563188  ...     -0.118124
632      21.766305  ...     -0.038763
633      19.862868  ...     -0.223210
634      17.229850  ...      0.027667
635      14.754098  ...     -0.313308

  pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change \
212      -0.099233      0.073848      -0.049592
213      0.227172      0.103821      0.096883
214      0.130662      -0.054713      0.024810
215      -0.126447      -0.252555      0.209940
216       0.746014      0.112869      -0.135661
..      ...
631       0.124117      0.041773      -0.045256
632       0.105594      0.257573      0.008721
633       0.270018      -0.215785      0.264911
634       0.084566      -0.018356      0.082177
635       0.337800      0.039733      -0.100402

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pct_3_person_change pct_4_or_more_person_change \
212 -0.137526 -0.160102
213 -0.124331 -0.346550
214 0.243424 0.114622
215 1.010748 -0.156068
216 -0.218806 0.529741
.. ...
631 0.184168 -0.090151
632 -0.235028 -0.037315
633 -0.079417 0.093299
634 -0.094960 -0.011590
635 0.156551 0.008477

household_median_income_1yr_change rent_change is_gentrified \
212 0.227048 0.319759 False
213 0.285714 0.287982 False
214 0.557427 0.345320 True
215 0.942861 0.227990 False
216 0.528947 0.199794 False
.. ...
631 0.305880 0.255121 False
632 0.269922 0.495868 True
633 0.214091 0.112299 False
634 0.180427 0.284653 False
635 0.211222 -1.000001 False

predictions
212 0.258370
213 0.221712
214 0.413787
215 0.205923
216 0.202203
.. ...
631 0.216205
632 0.459594
633 0.201650
634 0.275015
635 0.243368

[359 rows x 38 columns],
'test': zip_code year less_than_9th_grade 9th_to_12th_grade \
0 10001 2023 2.112304 5.034325
1 10002 2023 13.788751 15.215928
2 10003 2023 2.982706 1.313485
3 10004 2023 0.531561 0.365449
4 10005 2023 0.887372 1.665529
.. ...
205 11436 2023 6.311237 9.230353
207 11691 2023 11.703595 10.241795
208 11692 2023 6.898884 7.379100
209 11693 2023 7.781227 8.177245
210 11694 2023 4.847583 4.899359

high_school_graduate some_college_no_degree associates_degree \
0 8.638444 9.514170 3.199261
1 17.723585 10.347033 3.970177
2 4.802430 6.170644 2.531195
3 1.661130 6.378738 0.166113
4 5.870307 6.484642 0.832765
.. ...
205 36.083519 17.974187 8.169471
207 27.801784 18.210444 8.761607
208 31.268177 18.369970 9.543456
209 27.689179 14.224553 6.379107
210 25.124587 16.652644 5.210019

bachelors_degree graduate_or_professional_degree \
0 41.471572 30.029924
1 26.542462 12.412064
2 44.729641 37.469899
3 42.624585 48.272425
4 50.552901 33.706485
.. ...
205 15.048314 7.182918
207 12.567804 10.712972
208 16.307068 10.233345
209 21.920154 13.828535
210 22.516342 20.749466

pct_children_teens_5-19 ... pct_middle_aged_35-59_change \
0 10.653736 ... 0.088749
1 11.974787 ... -0.077519
2 17.103576 ... -0.126590

```

```

3          11.561290 ...          0.145315
4          5.877896 ...          0.024834
..          ... ...          ...
205        18.910316 ...         -0.077443
207        22.684269 ...         -0.093891
208        24.222480 ...         -0.106917
209        18.421858 ...         -0.293290
210        18.987401 ...          0.019953

pct_older_adults_60+_change  pct_1_person_change  pct_2_person_change  \
0          -0.058638          0.026323          -0.000095
1           0.193148          0.089236          0.118224
2           0.142495          -0.052623          0.010979
3          -0.156268          -0.194850          0.159015
4           2.030959          0.134394          -0.130185
..          ...          ...          ...
205         0.294014          0.081634          0.046443
207         0.248278          0.115528          -0.019349
208         0.198880          0.320081          -0.088869
209         0.496218          -0.100137          0.237027
210         0.068398          -0.051488          0.043743

pct_3_person_change  pct_4_or_more_person_change  \
0          -0.075879          -0.092145
1          -0.130464          -0.343500
2           0.398857          -0.054939
3           0.690141          -0.057121
4          -0.426038          0.827429
..          ...          ...
205        -0.028139          -0.046383
207         0.071431          -0.117662
208        -0.277919          0.049429
209        -0.174290          -0.004850
210         0.065194          -0.027958

household_median_income_lyr_change  rent_change  is_gentrified  \
0           0.421562          0.520362          True
1           0.379499          0.368481          True
2           0.566464          0.365517          True
3           0.843012          0.227990          False
4           0.705477          0.199794          False
..          ...          ...          ...
205         0.345270          0.464005          True
207         0.357043          0.344507          True
208         0.175185          0.504132          True
209         0.273043          0.281283          False
210         0.382161          0.474422          True

predictions
0          0.235080
1          0.221712
2          0.413787
3          0.205923
4          0.202203
..          ...
205         0.380048
207         0.205146
208         0.360961
209         0.217469
210         0.359467

[178 rows x 38 columns],
'perf': 0.7112388132701446}]

```

Crime Data Sub-Model

```

In [ ]: rents = pd.melt(rent_change_pivot.reset_index(), id_vars='zip_code', value_name='rent_change')
crime_data = pd.read_csv('datasets/nyc_open_datasets/crime_data_311_final.csv') #use snigdha's dataset...

feats = crime_data.columns[3:]

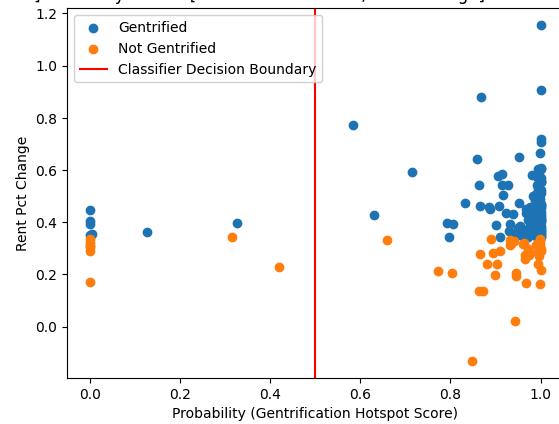
crime_data = crime_data.loc[crime_data['Year']>2015]
crime_data['zip_code'] = crime_data['Incident Zip'].astype(str)
crime_data['year'] = crime_data['Year'].astype(str)
combined_crime = crime_data.merge(
    rents,
    how='inner',
    on=['zip_code', 'year']
)
combined_crime['is_gentrified'] = combined_crime['rent_change']>0.343

train = combined_crime.loc[combined_crime['year'].astype(int).between(2021,2022)].dropna()
test = combined_crime.loc[combined_crime['year'].astype(int) == 2023].dropna()

```

```
crime_models = run_all_models(train, test, feats, target='is_gentrified')
crime_models
```

[BASELINE] NaiveBayes: Corr[Gentrification Score, Rent Change]=0.1513659097954626



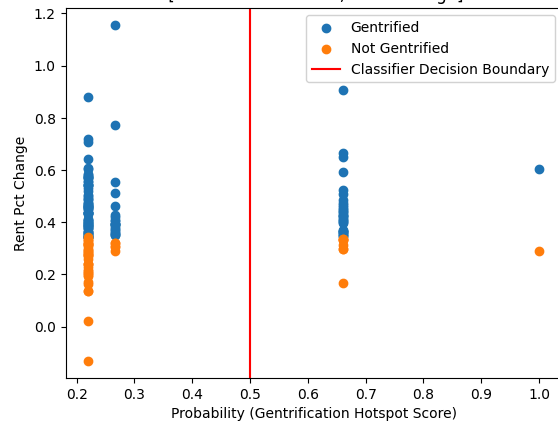
*** [BASELINE] NaiveBayes Accuracy: 0.7231638418079096 ***

*** [BASELINE] NaiveBayes F1: 0.8304498269896193 ***

*** Feature Importance Scores ***

{}

Single Decision Tree: Corr[Gentrification Score, Rent Change]=0.07741963493072607



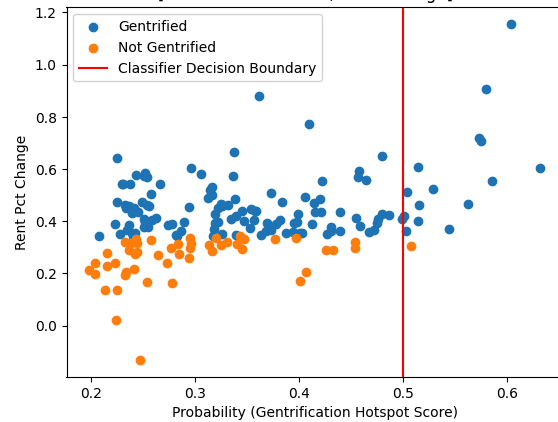
*** Single Decision Tree Accuracy: 0.4180790960451977 ***

*** Single Decision Tree F1: 0.3905325443786982 ***

*** Feature Importance Scores ***

{'Blocked Sidewalk': 0.27365941314375597, 'Mass Gathering': 0.2649752556181178, 'Grand Total': 0.19661643356150868, 'In Public': 0.19060423673050414, 'Tortured': 0.07414466094611355, 'Drag Racing': 0.0, 'Disruptive Passenger': 0.0, 'Congestion/Gridlock': 0.0, 'Commercial Overnight Parking': 0.0, 'Chronic Stoplight Violation': 0.0, 'Car/Truck Music': 0.0, 'Homeless Issue': 0.0, 'Car/Truck Horn': 0.0, 'Business not in compliance': 0.0, 'Broken Sidewalk': 0.0, 'Blocked Bike Lane': 0.0, 'After Hours - Licensed Est': 0.0, 'Grass/Weeds': 0.0, 'Truck Route Violation': 0.0, 'Hours of Operation': 0.0}

Random Forests: Corr[Gentrification Score, Rent Change]=0.4054466040252078



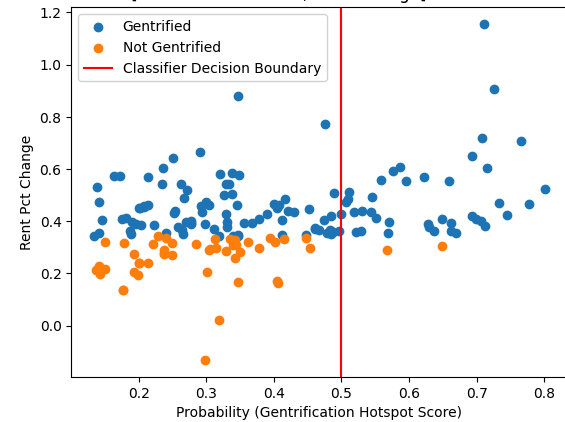
*** Random Forests Accuracy: 0.3615819209039548 ***

*** Random Forests F1: 0.2097902097902098 ***

*** Feature Importance Scores ***

{'Blocked Bike Lane': 0.09589412810259353, 'Blocked Sidewalk': 0.08605210963227036, 'Loud Talking': 0.07843501786478639, 'Commercial Overnight Parking': 0.07599080216990398, 'Grand Total': 0.06645240347079054, 'In Public': 0.06152479593788273, 'Congestion/Gridlock': 0.05879125544513605, 'Tortured': 0.05860896283398991, 'In Car': 0.0500332688764323, 'Loud Music/Party': 0.04916505001176793, 'Drag Racing': 0.04632298216338246, 'Car/Truck Music': 0.04481125947227659, 'Trespassing': 0.035508549239566395, 'Mass Gathering': 0.03452033706966171, 'Car/Truck Horn': 0.03309765453923891, 'Loud Television': 0.031682647032963675, 'Truck Route Violation': 0.0275325767727688, 'Business not in compliance': 0.01735355191061049, 'Chronic Stoplight Violation': 0.017060796091892306, 'After Hours - Licensed Est': 0.011435972805943627}

XGBoost: Corr[Gentrification Score, Rent Change]=0.3476731188678547



*** XGBoost Accuracy: 0.4915254237288136 ***

*** XGBoost F1: 0.4642857142857143 ***

*** Feature Importance Scores ***

{'In Public': 0.08000004, 'Car/Truck Music': 0.07770348, 'After Hours - Licensed Est': 0.07188516, 'Mass Gathering': 0.07171762, 'Commercial Overnight Parking': 0.060880587, 'Grand Total': 0.05970092, 'In Car': 0.052689467, 'Congestion/Gridlock': 0.052655254, 'Drag Racing': 0.052044656, 'Loud Talking': 0.048238456, 'Blocked Bike Lane': 0.046827327, 'Blocked Sidewalk': 0.043842107, 'Business not in compliance': 0.043377135, 'Loud Television': 0.036476027, 'Chronic Stoplight Violation': 0.035674386, 'Tortured': 0.03232693, 'Street Con Game': 0.03131353, 'Loud Music/Party': 0.028021716, 'Truck Route Violation': 0.027494173, 'Trespassing': 0.023991294}

```

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'Commercial Overnight Parking', 'Congestion/Gridlock',
'Disruptive Passenger', 'Drag Racing', 'Grass/Weeds', 'Homeless Issue',
'Hours of Operation', 'In Car', 'In Public', 'Lighting',
'Loud Music/Party', 'Loud Talking', 'Loud Television', 'Mass Gathering',
'Nuisance/Truant', 'People Created Noise', 'Performance',
'Sidewalk Violation', 'Street Con Game', 'Structure - Outdoors',
'Ticket Scalping', 'Tortured', 'Trespassing', 'Truck Route Violation'],
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'feature_importances': {},
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899  2021      10001      False      5332
900  2021      10002      False     14330
901  2021      10003      False      8537
902  2021      10004      False       647
903  2021      10005      False       644
...    ...      ...      ...      ...
1253 2022      11436       True      3705
1254 2022      11691      False     4832
1255 2022      11692       True      1567
1256 2022      11693      False      1879
1257 2022      11694       True      3802

      After Hours - Licensed Est Blocked Bike Lane Blocked Sidewalk \
899              3              217              80
900              2              594              281
901              2              348              110
902              0               2              26
903              0               8              34
...              ...              ...              ...
1253              0               0              81
1254              0               7              145
1255              0               7              40
1256              0              28              74
1257              0              17              162

      Broken Sidewalk Business not in compliance Car/Truck Horn ... \
899              0              0              34 ...
900              0              0              129 ...
901              0              0              36 ...
902              0              0               8 ...
903              0              0              19 ...
...              ...              ...              ...
1253              0              0               4 ...
1254              0              0              24 ...
1255              0              0               9 ...
1256              0              0               9 ...
1257              0              0              13 ...

      Street Con Game Structure - Outdoors Ticket Scalping Tortured \
899              3              0              0      8
900              0              0              0     15
901              0              0              0      0
902              0              0              0      0
903              0              0              0      2
...              ...              ...              ...
1253              0              0              0      1
1254              0              0              0      4
1255              0              0              0      4
1256              0              0              0      0
1257              0              0              0      0

      Trespassing Truck Route Violation zip_code year rent_change \
899             14              1  10001  2021  0.140774
900             86              2  10002  2021  0.187075
901             26              0  10003  2021  0.220690
902              0              0  10004  2021  0.227990
903              1              0  10005  2021  0.199794
...              ...              ...      ...
1253             4              1  11436  2022  0.421127
1254            23              0  11691  2022  0.325885
1255             6              1  11692  2022  0.540732
1256             1              2  11693  2022  0.216043
1257             8              1  11694  2022  0.392739

      predictions
899      0.000055

```

```

900      0.999618
901      0.999026
902      0.762090
903      0.874923
...      ...
1253     0.962761
1254     0.992729
1255     0.883279
1256     0.913343
1257     0.986133

```

```

[357 rows x 40 columns],
'test':
Year Incident Zip is_gentrified Grand Total \
1259 2023      10001      True      5522
1260 2023      10002      True     13198
1261 2023      10003      True     11781
1262 2023      10004     False      1327
1263 2023      10005     False       774
...      ...      ...      ...
1433 2023      11436      True      4313
1434 2023      11691      True      5507
1435 2023      11692      True      1876
1436 2023      11693     False      2234
1437 2023      11694      True      3876

```

```

After Hours - Licensed Est Blocked Bike Lane Blocked Sidewalk \
1259      0      172      173
1260      1     1235     201
1261      1      619     141
1262      0       8      90
1263      0      22     100
...      ...      ...
1433      0       1      90
1434      0      30     159
1435      0       1      54
1436      0      30      71
1437      0      15     138

```

```

Broken Sidewalk Business not in compliance Car/Truck Horn ... \
1259      0      0      82 ...
1260      0      0      99 ...
1261      0      0      50 ...
1262      0      0      10 ...
1263      0      0      15 ...
...      ...      ...
1433      0      0       3 ...
1434      0      0       2 ...
1435      0      0       1 ...
1436      0      0       4 ...
1437      0      0       9 ...

```

```

Street Con Game Structure - Outdoors Ticket Scalping Tortured \
1259      0      0      1      8
1260      0      0      0      9
1261      0      0      0      5
1262      0      0      2      0
1263      0      0      0      0
...      ...      ...
1433      1      0      0      7
1434      0      0      0      2
1435      0      0      0      4
1436      0      0      0      5
1437      0      0      0      1

```

```

Trespassing Truck Route Violation zip_code year rent_change \
1259      21      1 10001 2023 0.520362
1260     140      1 10002 2023 0.368481
1261      30      1 10003 2023 0.365517
1262       0      1 10004 2023 0.227990
1263       2      0 10005 2023 0.199794
...      ...      ...
1433       6      0 11436 2023 0.464085
1434      25      0 11691 2023 0.344507
1435       2      0 11692 2023 0.504132
1436       7      0 11693 2023 0.281283
1437      33      0 11694 2023 0.474422

```

```

predictions
1259     0.997657
1260     1.000000
1261     0.999999
1262     0.421037
1263     0.898868
...      ...

```

```

1433      0.908546
1434      0.973000
1435      0.917588
1436      0.893935
1437      0.952297

[177 rows x 40 columns],
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{'model': DecisionTreeClassifier(max_depth=3, min_samples_split=5),
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 'Car/Truck Horn', 'Car/Truck Music', 'Chronic Stoplight Violation',
 'Commercial Overnight Parking', 'Congestion/Gridlock',
 'Disruptive Passenger', 'Drag Racing', 'Grass/Weeds', 'Homeless Issue',
 'Hours of Operation', 'In Car', 'In Public', 'Lighting',
 'Loud Music/Party', 'Loud Talking', 'Loud Television', 'Mass Gathering',
 'Nuisance/Truant', 'People Created Noise', 'Performance',
 'Sidewalk Violation', 'Street Con Game', 'Structure - Outdoors',
 'Ticket Scalping', 'Tortured', 'Trespassing', 'Truck Route Violation'],
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 'Mass Gathering': 0.2649752556181178,
 'Grand Total': 0.19661643356150868,
 'In Public': 0.19060423673050414,
 'Tortured': 0.07414466094611355,
 'Drag Racing': 0.0,
 'Disruptive Passenger': 0.0,
 'Congestion/Gridlock': 0.0,
 'Commercial Overnight Parking': 0.0,
 'Chronic Stoplight Violation': 0.0,
 'Car/Truck Music': 0.0,
 'Homeless Issue': 0.0,
 'Car/Truck Horn': 0.0,
 'Business not in compliance': 0.0,
 'Broken Sidewalk': 0.0,
 'Blocked Bike Lane': 0.0,
 'After Hours - Licensed Est': 0.0,
 'Grass/Weeds': 0.0,
 'Truck Route Violation': 0.0,
 'Hours of Operation': 0.0},
 'train':
      Year  Incident Zip  is_gentrified  Grand Total \
899  2021      10001      False          5332
900  2021      10002      False          14330
901  2021      10003      False          8537
902  2021      10004      False           647
903  2021      10005      False           644
...  ...      ...      ...      ...
1253 2022      11436      True          3705
1254 2022      11691      False          4832
1255 2022      11692      True          1567
1256 2022      11693      False          1879
1257 2022      11694      True          3802

      After Hours - Licensed Est  Blocked Bike Lane  Blocked Sidewalk \
899                          3                217             80
900                          2                594            281
901                          2                348            110
902                          0                 2             26
903                          0                 8             34
...                          ...                ...            ...
1253                         0                 0             81
1254                         0                 7            145
1255                         0                 7             40
1256                         0                28             74
1257                         0                17            162

      Broken Sidewalk  Business not in compliance  Car/Truck Horn  ... \
899                  0                        0                34  ...
900                  0                        0               129  ...
901                  0                        0                36  ...
902                  0                        0                 8  ...
903                  0                        0                19  ...
...                  ...                      ...              ...
1253                  0                        0                 4  ...
1254                  0                        0                24  ...
1255                  0                        0                 9  ...
1256                  0                        0                 9  ...
1257                  0                        0                13  ...

```

	Street Con Game	Structure - Outdoors	Ticket Scalping	Tortured \
899	3	0	0	8
900	0	0	0	15
901	0	0	0	0
902	0	0	0	0
903	0	0	0	2
...	...	...	...	...
1253	0	0	0	1
1254	0	0	0	4
1255	0	0	0	4
1256	0	0	0	0
1257	0	0	0	0

	Trespassing	Truck Route Violation	zip_code	year	rent_change \
899	14	1	10001	2021	0.140774
900	86	2	10002	2021	0.187075
901	26	0	10003	2021	0.220690
902	0	0	10004	2021	0.227990
903	1	0	10005	2021	0.199794
...	...	...	...	...	...
1253	4	1	11436	2022	0.421127
1254	23	0	11691	2022	0.325885
1255	6	1	11692	2022	0.540732
1256	1	2	11693	2022	0.216043
1257	8	1	11694	2022	0.392739

	predictions
899	0.219828
900	0.266667
901	0.219828
902	0.219828
903	0.219828
...	...
1253	0.219828
1254	0.219828
1255	0.219828
1256	0.219828
1257	0.661538

[357 rows x 40 columns],				
'test':	Year	Incident Zip	is_gentrified	Grand Total \
1259	2023	10001	True	5522
1260	2023	10002	True	13198
1261	2023	10003	True	11781
1262	2023	10004	False	1327
1263	2023	10005	False	774
...	...	...	...	...
1433	2023	11436	True	4313
1434	2023	11691	True	5507
1435	2023	11692	True	1876
1436	2023	11693	False	2234
1437	2023	11694	True	3876

	After Hours - Licensed Est	Blocked Bike Lane	Blocked Sidewalk \
1259	0	172	173
1260	1	1235	201
1261	1	619	141
1262	0	8	90
1263	0	22	100
...	...	...	...
1433	0	1	90
1434	0	30	159
1435	0	1	54
1436	0	30	71
1437	0	15	138

	Broken Sidewalk	Business not in compliance	Car/Truck Horn ... \
1259	0	0	82 ...
1260	0	0	99 ...
1261	0	0	50 ...
1262	0	0	10 ...
1263	0	0	15 ...
...	...	...	...
1433	0	0	3 ...
1434	0	0	2 ...
1435	0	0	1 ...
1436	0	0	4 ...
1437	0	0	9 ...

	Street Con Game	Structure - Outdoors	Ticket Scalping	Tortured \
1259	0	0	1	8
1260	0	0	0	9
1261	0	0	0	5

```

1262      0      0      2      0
1263      0      0      0      0
...      ...      ...      ...      ...
1433      1      0      0      7
1434      0      0      0      2
1435      0      0      0      4
1436      0      0      0      5
1437      0      0      0      1

Trespassing Truck Route Violation zip_code year rent_change \
1259      21      1      10001      2023      0.520362
1260      140      1      10002      2023      0.368481
1261      30      1      10003      2023      0.365517
1262      0      1      10004      2023      0.227990
1263      2      0      10005      2023      0.199794
...      ...      ...      ...      ...
1433      6      2      11436      2023      0.464085
1434      25      0      11691      2023      0.344507
1435      2      0      11692      2023      0.504132
1436      7      0      11693      2023      0.281283
1437      33      0      11694      2023      0.474422

predictions
1259      0.219828
1260      0.266667
1261      0.661538
1262      0.219828
1263      0.219828
...      ...
1433      0.219828
1434      0.219828
1435      0.219828
1436      0.219828
1437      0.219828

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'min_samples_leaf': [1, 2, 5]},
'nm': 'Random Forests',
'training_feats': Index(['Grand Total', 'After Hours - Licensed Est', 'Blocked Bike Lane',
'Blocked Sidewalk', 'Broken Sidewalk', 'Business not in compliance',
'Car/Truck Horn', 'Car/Truck Music', 'Chronic Stoplight Violation',
'Commercial Overnight Parking', 'Congestion/Gridlock',
'Disruptive Passenger', 'Drag Racing', 'Grass/Weeds', 'Homeless Issue',
'Hours of Operation', 'In Car', 'In Public', 'Lighting',
'Loud Music/Party', 'Loud Talking', 'Loud Television', 'Mass Gathering',
'Nuisance/Truant', 'People Created Noise', 'Performance',
'Sidewalk Violation', 'Street Con Game', 'Structure - Outdoors',
'Ticket Scalping', 'Tortured', 'Trespassing', 'Truck Route Violation'],
dtype='object'),
'prediction_var': 'is_gentrified',
'plot': True,
'feature_importances': {'Blocked Bike Lane': 0.09589412810259353,
'Blocked Sidewalk': 0.08605210963227036,
'Loud Talking': 0.07843501786478639,
'Commercial Overnight Parking': 0.07599080216990398,
'Grand Total': 0.06645240347079054,
'In Public': 0.06152479593788273,
'Congestion/Gridlock': 0.05879125544513605,
'Tortured': 0.05860896283398991,
'In Car': 0.05003332688764323,
'Loud Music/Party': 0.04916505001176793,
'Drag Racing': 0.04632298216338246,
'Car/Truck Music': 0.04481125947227659,
'Trespassing': 0.035508549239566395,
'Mass Gathering': 0.03452033706966171,
'Car/Truck Horn': 0.03309765453923891,
'Loud Television': 0.031682647032963675,
'Truck Route Violation': 0.0275325767727688,
'Business not in compliance': 0.01735355191061049,
'Chronic Stoplight Violation': 0.017060796091892306,
'After Hours - Licensed Est': 0.011435972805943627},
'train':      Year Incident Zip is_gentrified Grand Total \
899      2021      10001      False      5332
900      2021      10002      False      14330
901      2021      10003      False      8537
902      2021      10004      False      647
903      2021      10005      False      644
...      ...      ...      ...      ...
1253      2022      11436      True      3705

```

1254	2022	11691	False	4832
1255	2022	11692	True	1567
1256	2022	11693	False	1879
1257	2022	11694	True	3802

	After Hours -	Licensed Est	Blocked Bike Lane	Blocked Sidewalk \
899		3	217	80
900		2	594	281
901		2	348	110
902		0	2	26
903		0	8	34
...	...	...	...	...
1253		0	0	81
1254		0	7	145
1255		0	7	40
1256		0	28	74
1257		0	17	162

	Broken Sidewalk	Business not in compliance	Car/Truck Horn ... \
899	0	0	34 ...
900	0	0	129 ...
901	0	0	36 ...
902	0	0	8 ...
903	0	0	19 ...
...	...	...	...
1253	0	0	4 ...
1254	0	0	24 ...
1255	0	0	9 ...
1256	0	0	9 ...
1257	0	0	13 ...

	Street Con Game	Structure - Outdoors	Ticket Scalping	Tortured \
899	3	0	0	8
900	0	0	0	15
901	0	0	0	0
902	0	0	0	0
903	0	0	0	2
...	...	...	...	...
1253	0	0	0	1
1254	0	0	0	4
1255	0	0	0	4
1256	0	0	0	0
1257	0	0	0	0

	Trespassing	Truck Route Violation	zip_code	year	rent_change \
899	14	1	10001	2021	0.140774
900	86	2	10002	2021	0.187075
901	26	0	10003	2021	0.220690
902	0	0	10004	2021	0.227990
903	1	0	10005	2021	0.199794
...	...	...	...	...	...
1253	4	1	11436	2022	0.421127
1254	23	0	11691	2022	0.325885
1255	6	1	11692	2022	0.540732
1256	1	2	11693	2022	0.216043
1257	8	1	11694	2022	0.392739

	predictions
899	0.277354
900	0.394824
901	0.330284
902	0.199290
903	0.195953
...	...
1253	0.244459
1254	0.273791
1255	0.258860
1256	0.216869
1257	0.305292

[357 rows x 40 columns],				
'test':	Year	Incident Zip	is_gentrified	Grand Total \
1259	2023	10001	True	5522
1260	2023	10002	True	13198
1261	2023	10003	True	11781
1262	2023	10004	False	1327
1263	2023	10005	False	774
...	...	...	...	...
1433	2023	11436	True	4313
1434	2023	11691	True	5507
1435	2023	11692	True	1876
1436	2023	11693	False	2234
1437	2023	11694	True	3876

```

After Hours - Licensed Est Blocked Bike Lane Blocked Sidewalk \
1259 0 172 173
1260 1 1235 201
1261 1 619 141
1262 0 8 90
1263 0 22 100
... ..
1433 0 1 90
1434 0 30 159
1435 0 1 54
1436 0 30 71
1437 0 15 138

Broken Sidewalk Business not in compliance Car/Truck Horn ... \
1259 0 0 82 ...
1260 0 0 99 ...
1261 0 0 50 ...
1262 0 0 10 ...
1263 0 0 15 ...
... ..
1433 0 0 3 ...
1434 0 0 2 ...
1435 0 0 1 ...
1436 0 0 4 ...
1437 0 0 9 ...

Street Con Game Structure - Outdoors Ticket Scalping Tortured \
1259 0 0 1 8
1260 0 0 0 9
1261 0 0 0 5
1262 0 0 2 0
1263 0 0 0 0
... ..
1433 1 0 0 7
1434 0 0 0 2
1435 0 0 0 4
1436 0 0 0 5
1437 0 0 0 1

Trespassing Truck Route Violation zip_code year rent_change \
1259 21 1 10001 2023 0.520362
1260 140 1 10002 2023 0.368481
1261 30 1 10003 2023 0.365517
1262 0 1 10004 2023 0.227990
1263 2 0 10005 2023 0.199794
... ..
1433 6 2 11436 2023 0.464085
1434 25 0 11691 2023 0.344507
1435 2 0 11692 2023 0.504132
1436 7 0 11693 2023 0.281283
1437 33 0 11694 2023 0.474422

predictions
1259 0.314515
1260 0.543869
1261 0.472333
1262 0.214602
1263 0.203178
... ..
1433 0.253828
1434 0.282372
1435 0.257108
1436 0.244298
1437 0.248375

[177 rows x 40 columns],
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    enable_categorical=False, eval_metric=None, feature_types=None,
    feature_weights=None, gamma=None, grow_policy=None,
    importance_type=None, interaction_constraints=None,
    learning_rate=0.01, max_bin=None, max_cat_threshold=None,
    max_cat_to_onehot=None, max_delta_step=None, max_depth=3,
    max_leaves=None, min_child_weight=None, missing=nan,
    monotone_constraints=None, multi_strategy=None, n_estimators=300,
    n_jobs=None, num_parallel_tree=None, ...),
'params_config': {'n_estimators': [50, 100, 200, 300],
'max_depth': [3, 5, 7, 9],
'learning_rate': [0.01, 0.05, 0.1, 0.2]},
'nm': 'XGBoost',
'training_feats': Index(['Grand Total', 'After Hours - Licensed Est', 'Blocked Bike Lane',
    'Blocked Sidewalk', 'Broken Sidewalk', 'Business not in compliance',

```

```

'Car/Truck Horn', 'Car/Truck Music', 'Chronic Stoplight Violation',
'Commercial Overnight Parking', 'Congestion/Gridlock',
'Disruptive Passenger', 'Drag Racing', 'Grass/Weeds', 'Homeless Issue',
'Hours of Operation', 'In Car', 'In Public', 'Lighting',
'Loud Music/Party', 'Loud Talking', 'Loud Television', 'Mass Gathering',
'Nuisance/Truant', 'People Created Noise', 'Performance',
'Sidewalk Violation', 'Street Con Game', 'Structure - Outdoors',
'Ticket Scalping', 'Tortured', 'Trespassing', 'Truck Route Violation'],
dtype='object'),
'prediction_var': 'is_gentrified',
'plot': True,
'feature_importances': {'In Public': 0.08000004,
'Car/Truck Music': 0.07770348,
'After Hours - Licensed Est': 0.07188516,
'Mass Gathering': 0.07171762,
'Commercial Overnight Parking': 0.060880587,
'Grand Total': 0.05970092,
'In Car': 0.052689467,
'Congestion/Gridlock': 0.052655254,
'Drag Racing': 0.052044656,
'Loud Talking': 0.048238456,
'Blocked Bike Lane': 0.046827327,
'Blocked Sidewalk': 0.043842107,
'Business not in compliance': 0.043377135,
'Loud Television': 0.036476027,
'Chronic Stoplight Violation': 0.035674386,
'Tortured': 0.03232693,
'Street Con Game': 0.03131353,
'Loud Music/Party': 0.028021716,
'Truck Route Violation': 0.027494173,
'Trespassing': 0.023991294},
'train':
Year Incident Zip is_gentrified Grand Total \
899 2021 10001 False 5332
900 2021 10002 False 14330
901 2021 10003 False 8537
902 2021 10004 False 647
903 2021 10005 False 644
... ...
1253 2022 11436 True 3705
1254 2022 11691 False 4832
1255 2022 11692 True 1567
1256 2022 11693 False 1879
1257 2022 11694 True 3802

After Hours - Licensed Est Blocked Bike Lane Blocked Sidewalk \
899 3 217 80
900 2 594 281
901 2 348 110
902 0 2 26
903 0 8 34
... ...
1253 0 0 81
1254 0 7 145
1255 0 7 40
1256 0 28 74
1257 0 17 162

Broken Sidewalk Business not in compliance Car/Truck Horn ... \
899 0 0 34 ...
900 0 0 129 ...
901 0 0 36 ...
902 0 0 8 ...
903 0 0 19 ...
... ...
1253 0 0 4 ...
1254 0 0 24 ...
1255 0 0 9 ...
1256 0 0 9 ...
1257 0 0 13 ...

Street Con Game Structure - Outdoors Ticket Scalping Tortured \
899 3 0 0 8
900 0 0 0 15
901 0 0 0 0
902 0 0 0 0
903 0 0 0 2
... ...
1253 0 0 0 1
1254 0 0 0 4
1255 0 0 0 4
1256 0 0 0 0
1257 0 0 0 0

Trespassing Truck Route Violation zip_code year rent_change \

```

899	14	1	10001	2021	0.140774
900	86	2	10002	2021	0.187075
901	26	0	10003	2021	0.220690
902	0	0	10004	2021	0.227990
903	1	0	10005	2021	0.199794
...	...	...	...	...	...
1253	4	1	11436	2022	0.421127
1254	23	0	11691	2022	0.325885
1255	6	1	11692	2022	0.540732
1256	1	2	11693	2022	0.216043
1257	8	1	11694	2022	0.392739

predictions

899	0.125669
900	0.258673
901	0.282606
902	0.160032
903	0.137672
...	...
1253	0.222838
1254	0.327961
1255	0.319909
1256	0.310036
1257	0.472079

[357 rows x 40 columns],

'test':

	Year	Incident Zip	is_gentrified	Grand Total	\
1259	2023	10001	True	5522	
1260	2023	10002	True	13198	
1261	2023	10003	True	11781	
1262	2023	10004	False	1327	
1263	2023	10005	False	774	
...	...	...	...	...	
1433	2023	11436	True	4313	
1434	2023	11691	True	5507	
1435	2023	11692	True	1876	
1436	2023	11693	False	2234	
1437	2023	11694	True	3876	

	After Hours -	Licensed Est	Blocked Bike Lane	Blocked Sidewalk	\
1259		0	172	173	
1260		1	1235	201	
1261		1	619	141	
1262		0	8	90	
1263		0	22	100	
...	...	...	...	...	
1433		0	1	90	
1434		0	30	159	
1435		0	1	54	
1436		0	30	71	
1437		0	15	138	

	Broken Sidewalk	Business not in compliance	Car/Truck Horn	...	\
1259	0	0	82	...	
1260	0	0	99	...	
1261	0	0	50	...	
1262	0	0	10	...	
1263	0	0	15	...	
...	...	...	...	...	
1433	0	0	3	...	
1434	0	0	2	...	
1435	0	0	1	...	
1436	0	0	4	...	
1437	0	0	9	...	

	Street Con Game	Structure - Outdoors	Ticket Scalping	Tortured	\
1259	0	0	1	8	
1260	0	0	0	9	
1261	0	0	0	5	
1262	0	0	2	0	
1263	0	0	0	0	
...	...	...	...	...	
1433	1	0	0	7	
1434	0	0	0	2	
1435	0	0	0	4	
1436	0	0	0	5	
1437	0	0	0	1	

	Trespassing	Truck Route Violation	zip_code	year	rent_change	\
1259	21	1	10001	2023	0.520362	
1260	140	1	10002	2023	0.368481	
1261	30	1	10003	2023	0.365517	
1262	0	1	10004	2023	0.227990	
1263	2	0	10005	2023	0.199794	

```

...      ...      ...      ...      ...
1433      6      2      11436      2023      0.464085
1434      25      0      11691      2023      0.344507
1435      2      0      11692      2023      0.504132
1436      7      0      11693      2023      0.281283
1437      33      0      11694      2023      0.474422

...      ...
1259      predictions
1259      0.271087
1260      0.461104
1261      0.467306
1262      0.142115
1263      0.142423

...      ...
1433      0.213288
1434      0.318299
1435      0.338489
1436      0.349991
1437      0.299880

[177 rows x 40 columns],
'perf': 0.3476731188678547}]

```

Ensemble Model

```

In [ ]: #generate some training data...

training = []
test = []

#take into acct. every model trained per dataset
models = {
    'business':business_models,
    'acs':acs_models,
    'crime':crime_models
}

for nm,m, m_config in models.items():

    #identify the model with the best calibration (i.e. greatest degree of linear fit btw prob. scores and rent index...)
    best_model = max(m_config,key=lambda x:x['perf'])

    if 'zip_code' not in best_model['train'].columns:
        best_model['train'] = best_model['train'].reset_index(names='zip_code')
    if 'zip_code' not in best_model['test'].columns:
        best_model['test'] = best_model['test'].reset_index(names='zip_code')

    #build ensemble feature for train/testing
    #aggregate pred. score results for training? mean, most recent score, etc.
    tr = best_model['train'].groupby('zip_code')[['predictions']].mean()\
        .rename(
            columns={
                'predictions':f"preds_{best_model['nm']}_{nm}"
            }
        )

    ts = best_model['test'].set_index('zip_code')[['predictions']]
        .rename(
            columns={
                'predictions':f"preds_{best_model['nm']}_{nm}"
            }
        )

    training.append(tr)
    test.append(ts)

#build training/test sets
train_TOTAL_ENSEMBLE = pd.concat(training,axis=1)
test_TOTAL_ENSEMBLE = pd.concat(test,axis=1)

#add target variable...
train_TOTAL_ENSEMBLE_w_rents = pd.concat((
    train_TOTAL_ENSEMBLE,
    pd.DataFrame(rent_change_pivot[['2022']].mean(axis=1),columns=['rent_change'])
),
    axis=1
)

test_TOTAL_ENSEMBLE_w_rents = pd.concat((
    test_TOTAL_ENSEMBLE,
    rent_change_pivot[['2023']].rename(
        columns={
            '2023':'rent_change'
        }
    )
),
    axis=1
)

```

```

    ),
    axis=1
)

#add binary outcome variable
train_TOTAL_ENSEMBLE_w_rents['is_gentrified'] = train_TOTAL_ENSEMBLE_w_rents['rent_change'] > 0.343
test_TOTAL_ENSEMBLE_w_rents['is_gentrified'] = test_TOTAL_ENSEMBLE_w_rents['rent_change'] > 0.343

train_TOTAL_ENSEMBLE_w_rents_f = train_TOTAL_ENSEMBLE_w_rents.dropna().drop(columns=['rent_change']).dropna()
test_TOTAL_ENSEMBLE_w_rents_f = test_TOTAL_ENSEMBLE_w_rents.dropna().drop(columns=['rent_change']).dropna()

```

```

In [ ]: from sklearn.linear_model import LogisticRegression
        from sklearn.preprocessing import StandardScaler

        #apply a logistic regression with increased penalization
        lr = LogisticRegression(penalty='l2', C=0.009)

        feats = train_TOTAL_ENSEMBLE_w_rents_f.columns[:3]

        #standardized sub-scores before fitting
        trainX, trainy = StandardScaler().fit_transform(train_TOTAL_ENSEMBLE_w_rents_f[feats].values), train_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']
        testX, testy = StandardScaler().fit_transform(test_TOTAL_ENSEMBLE_w_rents_f[feats].values), test_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']

        lr.fit(trainX, trainy)
        coefs = dict(zip(feats, lr.coef_[0]))

        #identify most sig. features via regression weights...
        coefs

```

```

Out[ ]: {'preds_XGBoost_business': 0.38930714388454335,
        'preds_XGBoost_acs': 0.2891655523756452,
        'preds_Random Forests_crime': 0.2264525411459326}

```

```

In [ ]: #binary outcome performance...
        preds = lr.predict(testX)
        test_TOTAL_ENSEMBLE_w_rents_f['predictions'] = lr.predict_proba(testX)[:,1]
        test_TOTAL_ENSEMBLE_w_rents_f['preds_standardized'] = np.average(testX, weights=list(coefs.values()), axis=1)
        f1_score(testy, preds)

```

```

Out[ ]: 0.7955801104972375

```

```

In [ ]: trainX.mean(axis=1)

```

```

Out[ ]: array([-0.87842387, -0.53000612,  0.48516972, -1.19566189, -1.17512446,
        -1.14459605, -1.17732006, -0.57795918, -1.04327599, -0.87509294,
        -1.01423084,  0.37094634, -1.0334866 , -0.83117696, -1.03762902,
        -0.83174031, -0.74356182, -0.19177169, -1.04652493, -1.10754981,
        0.69592487, -0.59031348,  1.53818963,  1.02419853, -0.89068034,
        -0.58499082,  0.74604281,  1.25555501,  0.41835319,  0.68949838,
        0.21572891,  1.01565697, -0.83692255,  0.14767606,  0.32757834,
        -0.82721891,  0.16196948, -1.09429408, -1.12518103, -1.00280289,
        -1.13144651, -1.06767351,  0.3207583 , -0.813822 , -1.01199896,
        -0.76163843,  0.79724381,  0.68725132,  0.32598355,  0.46408207,
        0.6462071 , -0.65406335, -0.91887252,  0.92258486,  1.03593295,
        -0.63986001, -1.01921442,  0.41895144,  0.64800852,  0.29529578,
        0.46504093,  0.67611367, -0.83941168, -0.78554989, -0.96475547,
        -0.58752797, -1.16410558, -0.80659484, -0.89830803,  1.88136718,
        0.98582446,  0.6166229 , -0.575774 ,  0.26252643,  0.62970274,
        1.65692471, -1.05874416,  0.52355773,  1.98255897,  1.73111599,
        0.68837101, -0.76100108, -0.56050454,  0.25380113,  1.90420943,
        -0.73213713,  1.02147284,  0.58203377,  1.13718229,  1.72821625,
        1.8202088 ,  0.92916233,  0.7619081 ,  1.61555559,  1.74262479,
        0.90508905, -0.81885062,  0.57595661,  1.03841003, -0.85531518,
        0.33593154,  0.95863534,  0.75538433,  1.15871777,  0.6887042 ,
        -0.63816338,  0.51063062, -0.99845905,  1.63963181,  2.07123521,
        -0.95819522, -0.86904255,  0.13954248,  0.28787651, -0.96540175,
        -0.97186941,  0.30361548,  0.31986917, -0.72834943, -0.65939591,
        0.74364567, -0.7771569 , -0.76847167, -0.74561181,  0.9708894 ,
        -0.75274167, -0.58367116,  0.54702176,  0.84898355,  0.91776634,
        -0.81356734, -0.72719902,  0.64164692, -0.8444843 , -0.93699641,
        -0.72705026, -0.81520986,  0.48566979])

```

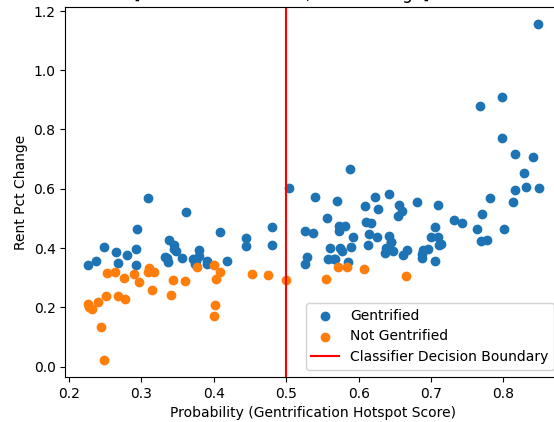
```

In [ ]: #probabilistic outcome performance...
        corr_ens = np.corrcoef(test_TOTAL_ENSEMBLE_w_rents_f['predictions'], test_TOTAL_ENSEMBLE_w_rents_f['rent_change'])[0][1]
        _, ax = plt.subplots()
        ax.scatter(
            test_TOTAL_ENSEMBLE_w_rents_f.loc[test_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']]['predictions'],
            test_TOTAL_ENSEMBLE_w_rents_f.loc[test_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']]['rent_change'],
            label='Gentrified'
        )
        ax.scatter(
            test_TOTAL_ENSEMBLE_w_rents_f.loc[~test_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']]['predictions'],
            test_TOTAL_ENSEMBLE_w_rents_f.loc[~test_TOTAL_ENSEMBLE_w_rents_f['is_gentrified']]['rent_change'],
            label='Not Gentrified'
        )

```

```
)
ax.axvline(0.5, label='Classifier Decision Boundary', color='red')
ax.set_title(f'ENSEMBLE: Corr[Gentrification Score, Rent Change]={corr_ens}')
ax.set_xlabel('Probability (Gentrification Hotspot Score)')
ax.set_ylabel('Rent Pct Change')
ax.legend()
plt.show()
```

ENSEMBLE: Corr[Gentrification Score, Rent Change]=0.6594197054908227



```
In [ ]: #plot the final scores spatially
w_geo = gpd.GeoDataFrame(pd.concat((geo_zip_nyc, test_TOTAL_ENSEMBLE_w_rents_f), axis=1).dropna(), crs="EPSG:4326")

fig = px.choropleth_mapbox(
    w_geo,
    geojson=w_geo['geometry'],
    locations=w_geo.index,
    color="predictions",
    featureidkey="id",
    center={"lat": w_geo.geometry.centroid.y.mean(), "lon": w_geo.geometry.centroid.x.mean()},
    zoom=10,
    mapbox_style="carto-positron",
)
fig.update_layout(margin={"r": 0, "t": 0, "l": 0, "b": 0})
#fig.show()
```

```
In [ ]: #plot the ground truth defn (rent index) spatially
fig = px.choropleth_mapbox(
    w_geo,
    geojson=w_geo['geometry'],
    locations=w_geo.index,
    color="rent_change",
    featureidkey="id",
    center={"lat": w_geo.geometry.centroid.y.mean(), "lon": w_geo.geometry.centroid.x.mean()},
    zoom=10,
    mapbox_style="carto-positron",
)
fig.update_layout(margin={"r": 0, "t": 0, "l": 0, "b": 0})
#fig.show()
```

```
In [ ]:
```